

Payoffs and perception mediate environmental feedback in an N -player trust game with Q -learning

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Abstract

Trust develops through learning, while collective behavior can alter the environment in which later decisions are made. Reinforcement-learning models describe adaptation, and eco-evolutionary models describe behavior-environment feedback, but how an endogenous environment changes trust through material incentives and perceived states remains unclear. We couple a fixed-role, two-population hierarchical trust game populated by heterogeneous tabular Q -learning agents to a centered endogenous environment. For the main feedback comparisons, we initialize the fixed-payoff baseline and feedback branches from the same learned state to control for prior learning. We then compare responses when payoff parameters or the observed environmental tier vary. As the payoff multiplication factor increased, successful trust in the fixed-payoff game crossed from a low to a higher finite-horizon level, led by risk-seeking investors. Environmental feedback promoted trust near perception-tier boundaries and reinforced its decline at the high-return operating point. At the high-return operating point, payoff feedback reinforced the learning-driven decline in trust. Stronger drive and longer horizons deepened this negative effect. Compared with a fixed-payoff trust game, environmental feedback carries past collective outcomes into future payoffs and observed states.

Keywords: Reinforcement learning, Trust game, Q -learning, Eco-evolutionary feedback, Finite-horizon dynamics, Behavioral heterogeneity

1. Introduction

Trust decisions expose individuals to uncertain returns from others' actions. The trust game introduced by Berg et al. [1] represents this problem through a sequential exchange. An investor decides whether to transfer resources, the transferred amount is multiplied, and a trustee decides how much to return. The investor risks exploitation, while the trustee can retain resources or reciprocate. A meta-analysis found that investors transferred about 50% of their endowment and trustees returned about 37% [2], departing from the self-interested one-shot equilibrium. Risk attitudes, time preferences, and prior experience also shape trust decisions [3, 4]. Repeated interaction adds learning and adaptation, which can change how trust develops over time. Trust also belongs to the broader study of human cooperation and moral preferences [5, 6, 7, 8, 9].

Evolutionary models of the N -player trust game and its replicator dynamics [10] examine how reputation, punishment, exclusion, and regulation sustain trust in structured populations [11, 12, 13, 14, 15, 16, 17]. Related trust-game studies examine diverse investment

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patterns [18], conditional investment in repeated groups [19], and fixed provider and consumer roles under different network structures [20]. These models typically use imitation, in which agents copy successful strategies. Reinforcement learning (RL) instead formalizes adaptation through an agent’s own trial-and-error experience [21]. Coupling RL with evolutionary games produces varied dynamics in cooperation dilemmas [22, 23, 24, 25, 26, 27, 28]. In trust games, Q -learning can generate high trust and trustworthiness [29], and reputation-coupled variants extend to networked and higher-order populations [30, 31].

Coevolutionary games couple strategic behavior to changes in the conditions of interaction [32]. Eco-evolutionary models also allow collective behavior to change the environment and hence payoffs, producing oscillations and tipping points [33, 34]. Trust-game models include margin-driven environmental feedback [35], loss assessment [36], and reputation-based nonlinear investment [37], within replicator or imitation dynamics. RL with environmental feedback has been studied chiefly in spatial public-goods games [27, 28]. In a hierarchical trust game, the environment can affect both the payoff received and the state on which a learned policy is conditioned. How these two routes shape trust remains unclear, particularly when the learning agents differ in their responses to gains and losses. Prospect theory provides a basis for representing this behavioral heterogeneity [38, 39].

Here, we study a two-population, fixed-role hierarchical trust game with tabular Q -learning, type-dependent rewards, and trustee discount factors. Collective outcomes drive an environmental state E_t , which maps to the payoff multiplication factor. For the main feedback comparisons, we initialize the fixed-payoff baseline and feedback branches from the same learned state. Each feedback branch follows its own environmental path.

Environmental feedback promoted trust near perception-tier boundaries but reinforced trust erosion at high-return operating points. At high-return operating points, payoff feedback reinforced the decline in trust driven by the learning process. One-round thresholds predict the participation ordering, while a discount-factor control shows how continuation value advances the learned onset. Together, these results show how past collective outcomes feed back through future payoffs and observed states. Perceived conditions can promote trust near a tier boundary, whereas payoff feedback can reinforce trust erosion at high returns.

Section 2 defines the model, centered feedback, and simulation protocol. Section 3 presents the baseline dynamics and paired feedback responses. Section 4 connects the mechanistic analysis with robustness checks, including type-specific behavior, threshold benchmarks, and finite-window outcome-pair dependence. Section 5 discusses the findings.

2. Model and methods

2.1. Populations and the trust game

We model investors $\mathcal{V}$ ($|\mathcal{V}| = N_I$) and trustees $\mathcal{T}$ ($|\mathcal{T}| = N_T$) with fixed roles and discrete-time interactions. Each round, investor i is matched uniformly at random to trustee $j(i, t)$, allowing many-to-one matching. The interaction follows the sequential trust-game protocol [1, 10]. Investor i chooses $a_i(t) \in A_I = \{0, 1\}$, corresponding to not investing or investing one unit of capital. Trustee j then chooses the trustworthy (T) or untrustworthy (U) action [10]. These choices determine the investor payoff,

$$r_i(t) = \begin{cases} m(t)/2 - 1, & a_i(t) = 1, a_j(t) = T, \\ -1, & a_i(t) = 1, a_j(t) = U, \\ 0, & a_i(t) = 0, \end{cases} \quad (1)$$

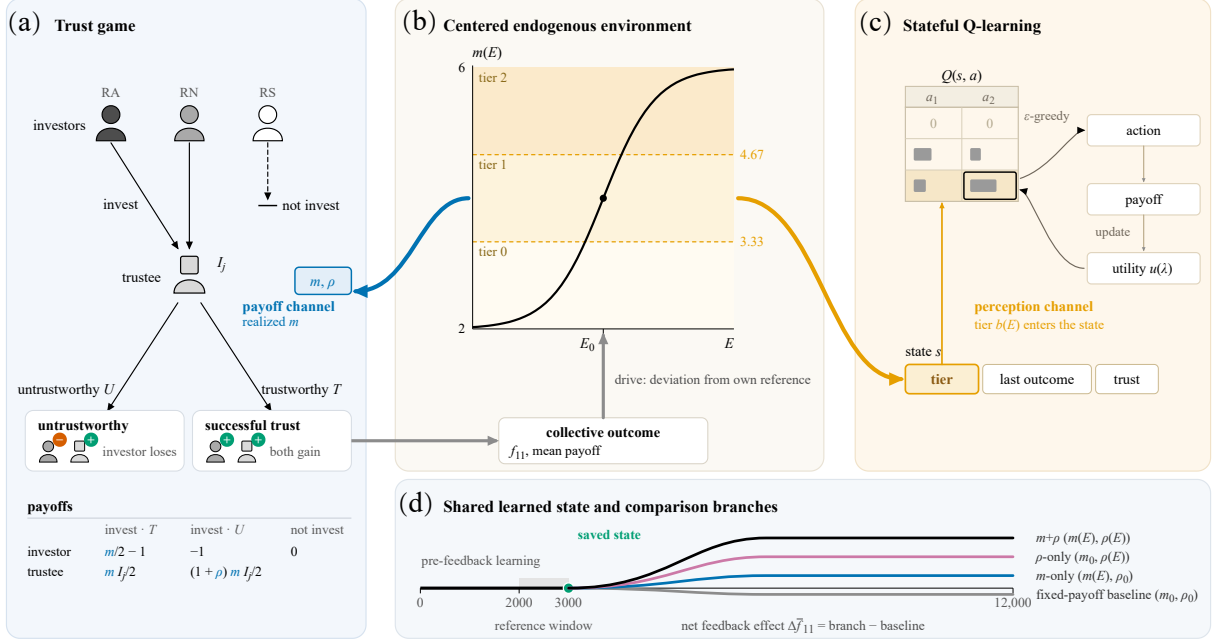


Figure 1: Model and paired comparisons with centered environmental feedback. (a) Investors choose whether to invest, matched trustees choose the trustworthy (\$T\$) or untrustworthy (\$U\$) action. (b) Collective outcomes update the environment, which enters realized payoffs through \$(m_t, \rho_t)\$ and decision states through the public tier \$b_E(t)\$. (c) Agents use tabular \$\epsilon\$-greedy \$Q\$-learning. (d) The learned state saved at round 3000 initializes the fixed-payoff baseline and the \$m\$-only, \$\rho\$-only and \$m+\rho\$ branches. Each branch uses the seed's reference over \$[2000, 3000)\$, feedback branches evolve their own environments.

and, with \$I_j(t) = \sum_{i \in \mathcal{M}_j(t)} a_i(t)\$ the number of matched investors who invest and \$\rho(t) \in (0, 1)\$ the defect ratio [40],

$$w_j(t) = \begin{cases} \frac{m(t)}{2} I_j(t), & a_j(t) = T, \\ \frac{(1 + \rho(t)) m(t)}{2} I_j(t), & a_j(t) = U. \end{cases} \quad (2)$$

The defect ratio raises the untrustworthy trustee's return by \$\rho(t)m(t)I_j(t)/2\$. Investor losses are defined separately in Eq. (1), so changing \$\rho\$ changes total surplus. Figure 1 summarizes one interaction round and the paired comparisons initialized from the same learned state at round 3000.

2.2. Q-learning with behavioral heterogeneity

Both populations use tabular \$Q\$-learning with \$\epsilon\$-greedy exploration and type-dependent reward functions. Investor types are labeled risk-averse, risk-neutral, and risk-seeking, \$\theta \in \{RA, RN, RS\}\$. The loss aversion coefficient sets the slope for negative payoffs, with \$\lambda_{RA} > 1 = \lambda_{RN} > \lambda_{RS}\$ in the prospect-theory-inspired value function [38]

$$u_\theta(r) = \begin{cases} r, & r \geq 0, \\ \lambda_\theta r, & r < 0. \end{cases} \quad (3)$$

We choose \$\lambda_{RA, RN, RS} = (2.0, 1.0, 0.5)\$ as contrasting model values for loss weights above, equal to, and below the unit gain weight. We assign trustees to long-sighted (\$L\$) and short-sighted (\$S\$) types whose utilities add a reputation term,

$$u_\phi(w_j, \Delta R_j) = w_j + \lambda_\phi \Delta R_j, \quad \lambda_L > \lambda_S, \quad \gamma_L > \gamma_S, \quad (4)$$

with $\Delta R_j := R_j(t+1) - R_j(t)$. Investor type enters only through the loss aversion coefficient λ_θ in Eq. (3), whereas trustee type enters through both the reputation weight λ_ϕ in Eq. (4) and the discount factor γ_ϕ in Eq. (6). Both actions remain available to every agent under ε -greedy selection.

Q -values follow standard temporal-difference updates,

$$Q_i(s_i, a_i) \leftarrow (1 - \alpha_Q)Q_i(s_i, a_i) + \alpha_Q[u_{\theta(i)}(r_i) + \gamma \max_{a'} Q_i(s'_i, a')], \quad (5)$$

$$Q_j(s_j, a_j) \leftarrow (1 - \alpha_Q)Q_j(s_j, a_j) + \alpha_Q[u_{\phi(j)}(w_j, \Delta R_j) + \gamma_\phi \max_{a'} Q_j(s'_j, a')]. \quad (6)$$

Reputations and pairwise trust weights evolve by exponential smoothing,

$$R_j(t+1) = (1 - \eta_R)R_j(t) + \eta_R \chi_j(t), \quad T_{ij}(t+1) = (1 - \eta_T)T_{ij}(t) + \eta_T \chi_{ij}(t), \quad (7)$$

with $\chi_j = \mathbf{1}[I_j > 0] \mathbf{1}[a_j = T]$ and $\chi_{ij} = \mathbf{1}[a_j = T]$ for matched invested pairs.

Investor states encode the last outcome, the matched trustee's discretized trust level, and the public environment tier ($|\mathcal{S}_I| = 4 \times 3 \times 3 = 36$). Trustee states encode own reputation, inflow, last action, average received trust, and the environment tier ($|\mathcal{S}_T| = 162$). All Q -tables start at zero, with uniform tie-breaking and exploration. Each binary action therefore has probability $1/2$ when an unupdated state row is first visited. Complete state definitions and binning functions are given in Appendix B.

2.3. Centered environment and the two channels

To represent feedback between collective behavior and its payoff environment [33, 34], we define a macroscopic state $E_t \in \mathbb{R}$ and a latent payoff-multiplication signal,

$$m_{\text{signal}}(E_t) = m_{\min} + (m_{\max} - m_{\min}) \sigma(kE_t), \quad (8)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

The environment affects agents through two channels. The *payoff channel* is the realized pair (m_t, ρ_t) entering Eqs. (1)–(2). The *perception channel* is the public environment tier $b_E(t) = b_3[(m_{\text{signal}}(E_t) - m_{\min})/(m_{\max} - m_{\min})]$, a component of every agent's decision state. Here b_3 is the three-level map defined in Appendix B. The tier boundaries lie at $m_{\text{signal}} = m_{\min} + (m_{\max} - m_{\min})/3 = 10/3 \simeq 3.333$ and $m_{\min} + 2(m_{\max} - m_{\min})/3 = 14/3 \simeq 4.667$. Entering a previously unvisited tier activates Q rows with no earlier updates; returning to a visited tier can reuse updated rows. The main payoff-channel branches apply this map to their own latent signals while differing in the payoff parameters they realize. Supplementary perception controls can instead hold the observed tier fixed.

For the dynamic- ρ controls, the defect-ratio tier is centered on the nominal environment,

$$b_\rho(E_t) = b_3[\sigma(k(E_t - E_0))], \quad (9)$$

$$\rho_{\text{dyn}}(E_t) = \rho_{b_\rho(E_t)},$$

$$(\rho_L, \rho_M, \rho_H) = (0.5, 1, 1.5) \times \rho_0,$$

so that $\rho_{\text{dyn}}(E_0) = \rho_0$ and, because the tiers of Eq. (B.1) are half-open, the tier leaves the middle band only when $E_t - E_0 < -\ln 2/k$ or $E_t - E_0 \geq \ln 2/k$ ($|E_t - E_0| \simeq 0.173$). We distinguish three payoff-channel interventions:

$$(m_t, \rho_t) = \begin{cases} (m_{\text{signal}}(E_t), \rho_0), & m\text{-only (main),} \\ (m_0, \rho_{\text{dyn}}(E_t)), & \rho\text{-only,} \\ (m_{\text{signal}}(E_t), \rho_{\text{dyn}}(E_t)), & m+\rho. \end{cases} \quad (10)$$

The ρ -only branch provides a perception control when its ρ -tier stays at the nominal value. Under this condition, both payoff parameters remain fixed, and the imposed change from the static baseline is the dynamic perceived state. Each feedback branch evolves its own E_t and perceived tier sequence. Where the ρ -tier changes, the ρ -only branch also changes realized payoff parameters. Hereafter, environment-feedback Q-learning (EF-Q) denotes the m -only main branch unless a channel is named explicitly.

The environment is driven by the volume-based cooperation rate $C_t := f_{11}(t)$ (the population share of successful-trust interactions, Section 2.4) and the population-mean investor payoff $\bar{r}_{I,t}$ (averaged over *all* investors, with non-investors contributing zero). Define

$$\begin{aligned} D_t &= \beta_C(C_t - C_{\text{ref}}) + \beta_r(\bar{r}_{I,t} - r_{\text{ref}}), \\ E_{t+1} &= E_0 + (1 - \nu)(E_t - E_0) + \nu D_t + \xi_t, \end{aligned} \tag{11}$$

where ν is the environmental update rate and ξ_t is optional Gaussian noise. The fixed anchor E_0 satisfies $m_{\text{signal}}(E_0) = m_0$. At the reference observables, $D_t = 0$, so a noiseless environment initialized at E_0 remains there. For constant D and zero noise, the environment approaches $E_0 + D$ with relaxation time $-1/\log(1 - \nu) \simeq 1/\nu$. The change in any one round is

$$E_{t+1} - E_t = \nu[D_t - (E_t - E_0)] + \xi_t.$$

Thus a positive drive can coexist with a falling environmental level. At fixed E_t and the other drive input, increasing either observable raises E_{t+1} and the next payoff-multiplication signal because both weights are positive. The loop-gain analysis below describes how behavioral responses to m modify this environmental relaxation.

For each nominal point and seed, we first run rounds 0–2999 with $(m_t, \rho_t) = (m_0, \rho_0)$ and $E_t = E_0$. This pre-feedback period visits only the nominal environment tier, leaving the Q rows for the other two tiers unupdated. We define the seed-specific reference values $(C_{\text{ref}}, r_{\text{ref}})$ as the time averages over rounds [2000, 3000). At round 3000, we save the complete learned state, memory variables, aggregate counters, and named random-number streams. We then clone this state to initialize the fixed-payoff baseline and the three feedback branches in Eq. (10). Because every branch starts from the same seed-specific learning history, their within-seed differences measure the finite-horizon effect of the imposed feedback, with the reference values held fixed.

2.4. Simulation settings and statistical methods

After round 3000, we hold $(m_t, \rho_t) = (m_0, \rho_0)$ fixed to define the fixed-payoff baseline. In the main EF-Q branch, we update E_t through Eq. (11), allow m_t to vary, and keep $\rho_t = \rho_0$. The ρ -only and $m+\rho$ branches isolate the additional channel changes defined above. The nominal anchor $E_0(m_0) = \sigma^{-1}[(m_0 - m_{\min})/(m_{\max} - m_{\min})]/k$ is finite only for $m_{\min} < m_0 < m_{\max}$. The 20×20 feedback grid uses cell centers, with $m_0 = 2.1, 2.3, \dots, 5.9$ and $\rho_0 = 0.06125, 0.08375, \dots, 0.48875$ (step 0.0225). All branch contrasts are paired within the same point-seed checkpoint. Algorithm 1 gives the simulation and paired-branch protocol.

We define the outcome state of each pairwise interaction as $s^{\text{out}} = (v_i, h_j)$, where $s^{\text{out}} \in \{00, 01, 10, 11\}$, with $v_i = a_i$ and $h_j = \mathbf{1}[a_j = T]$; $f_{ab}(t)$ denotes the population share of state ab . We call $f_{11}(t)$ the successful trust frequency. With $\mathcal{W} = \{3000, \dots, 11999\}$ the specified analysis window and S independent seeds, we define

$$\bar{f}_{ab} := \frac{1}{S} \sum_{s=1}^S \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} f_{ab}^{(s)}(t). \tag{12}$$

Algorithm 1: Environment-feedback Q -learning in the trust game

Input: Nominal parameters (m_0, ρ_0) , random seed, and model parameters in Table 1.

Output: Successful-trust trajectories, window statistics, and paired feedback effects.

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1 Assign investor and trustee types; set  $Q_i = Q_j = 0$ ,  $R_j = T_{ij} = 0.5$ , and  $E = E_0$ 
2 Set  $X_i(-1) = 00$  and  $a_j(-1) = U$ ; start with the static baseline
3 for  $t = 0, \dots, T_{\max} - 1$  do
4   if  $t = 3000$  then
5     Compute  $(C_{\text{ref}}, r_{\text{ref}})$  from rounds  $[2000, 3000)$  and hold them fixed
6     Clone the complete state, previous transitions, and random-number streams into
       the baseline,  $m$ -only,  $\rho$ -only, and  $m+\rho$  branches
7   foreach current branch do
8     Set  $(m_t, \rho_t) = (m_0, \rho_0)$  in the baseline; otherwise use Eq. (10). Compute  $b_E(t)$  using
       Eq. (B.2)
9     Match each investor  $i$  uniformly to a trustee  $j(i, t)$ 
10    Construct all investor states  $s_i(t)$  using Eq. (B.3)
11    if  $t > 0$  then update each investor's previous transition by Eq. (5), with  $s'_i = s_i(t)$ 
12    Select each  $a_i(t)$  by  $\varepsilon$ -greedy exploration with uniform tie-breaking
13    Aggregate  $I_j(t)$  and construct all trustee states  $s_j(t)$  (Eqs. (B.4) and (B.5))
14    if  $t > 0$  then update each trustee's previous transition by Eq. (6), with  $s'_j = s_j(t)$ 
15    Select each  $a_j(t)$  by the same exploration rule, including trustees with  $I_j(t) = 0$ 
16    Compute payoffs  $r_i(t)$  and  $w_j(t)$  using Eqs. (1)–(2)
17    Update  $R_j$ ,  $T_{ij}$ , and  $\Delta R_j$  using Eqs. (B.6)–(B.7)
18    Compute subjective rewards using Eqs. (3)–(4); retain each agent's current state,
       action, and reward for the next  $Q$  update
19    Record  $X_i(t)$ ,  $C_t$ ,  $\bar{r}_{I,t}$ , and  $f_{11}(t)$ 
20    if this is a feedback branch then
21      | Update  $E_{t+1}$  using Eq. (11)
22    else
23      | Keep  $E_{t+1} = E_0$ 
24 Compute window statistics and within-seed feedback-minus-baseline contrasts

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The primary estimand is the net feedback effect $\Delta \bar{f}_{11}^M := \bar{f}_{11}^{m\text{-only}} - \bar{f}_{11}^{\text{base}}$, computed as the within-seed paired change between the m -only branch and the static baseline; channel-control contrasts are defined analogously. We also record the baseline learning drift and realized environmental excursion for each seed. Baseline learning drift is the change in successful trust generated by continued learning under fixed payoff parameters, $\delta^{(s)} := \langle f_{11}^{(s)} \rangle_{\mathcal{W}} - \langle f_{11}^{(s)} \rangle_{[2000, 3000)}$. Economically, it measures endogenous erosion or recovery of trust under unchanged material incentives and separates ongoing adaptation from the additional effect of environmental feedback. The realized excursion is the window mean of the realized payoff multiplication factor minus its nominal value, $\Delta \bar{m}^{(s)} := \langle m_t^{(s)} \rangle_{\mathcal{W}} - m_0$.

At each ρ , we fit the fixed-horizon response $\bar{f}_{11}(m)$ with

$$f(m) = \frac{A}{1 + \exp[-(m - m^*)/w]} + c. \quad (13)$$

We report the fitted midpoint m^* and crossover band $m^* \pm w$. The cross-seed variance peak summarizes heterogeneity over the same window.

Over $\mathcal{W}$, we define $P_o(s, s')$ as the empirical joint frequency of consecutive outcome-state pairs and $P_e(s, s') := \pi(s)\pi(s')$ as the shuffled independence benchmark built from the

pooled same-window state frequencies. Following the cellwise normalization of Li et al. [24], we compute the normalized excess transition index (NETI),

$$\tilde{\kappa}(s, s') := \frac{P_o(s, s') - P_e(s, s')}{1 - P_e(s, s')} \quad (14)$$

which measures excess pair frequency with a chance correction analogous to Cohen’s κ [41]. A positive diagonal entry $\tilde{\kappa}(s, s)$ indicates self-persistence relative to the shuffled same-window benchmark. A positive off-diagonal entry identifies an over-represented directed outcome pair. Pooling across agents mixes within-agent sequence dependence, heterogeneity in individual action rates, and slow drift within $\mathcal{W}$. NETI summarizes these pooled dependencies. Its relation to conditional transition probabilities and its feasible bounds are derived in Appendix A.1.

In the main runs, we simulate 240 investors and 120 trustees for 12,000 rounds, with a burn-in cutoff at round 3,000. Table 1 lists the baseline model and simulation parameters; control-specific changes accompany each experiment. The main parameter and control runs use 20 seeds, the A–D comparisons use 50. We use 10 seeds to examine longer-term baseline dynamics (Fig. S1). With learning and exploration rates held constant, we analyze the specified window $\mathcal{W}$. We extended the simulations to 36,000 rounds and found continued trust decline at two parameter points and a late recovery at the third (Appendix D). Unless noted, error bands are two-sided pointwise 95% Student- t confidence intervals across the stated seed units. Bootstrap intervals are identified separately, and seed-level Pearson correlations describe association across matched seeds. The per-figure run specifications are listed in Appendix C.

Table 1: Baseline model and simulation parameters. Control-specific changes are stated with each experiment.

Group	Symbol	Value	Meaning
Population	N_I, N_T	240, 120	investors, trustees (main runs)
Learning	α_Q	0.1	learning rate (both roles)
	γ	0.9	investor discount factor
	γ_L, γ_S	0.95, 0.80	trustee discount factors
	ε	0.05	exploration rate (constant)
Types	$\lambda_{RA, RN, RS}$	(2.0, 1.0, 0.5)	loss aversion coefficients
	$\lambda_{L, S}$	(1.0, 0.1)	trustee reputation weights
	fractions	1/3 each; 1/2 each	type mixtures
Reputation	η_R, η_T	0.05, 0.05	reputation and trust smoothing
	R_0, T_0	0.5, 0.5	initial values; $Q \equiv 0$
Discretization	κ_1, κ_2	1/3, 2/3	three-tier binning; tier edges at $m = 3.333, 4.667$
Environment	ν	0.03	adjustment speed
	(β_C, β_r)	(1.4, 1.2)	drive weights
	k	4.0	sigmoid slope
	$(m_{\min}, m_{\max})$	(2, 6)	m range
	ρ tiers	$(0.5, 1, 1.5) \times \rho_0$	centered relative tiers
	σ_ξ	0 (main); {0.002, 0.005}	noise (robustness)
	$E_0(m)$	$\sigma^{-1}(\frac{m-2}{4})/k$	nominal environmental anchor
Runs	$T_{\max}$, burn-in cutoff	12000, 3000	fixed-horizon preset
	seeds	20 main; 50 A–D; 10 S1	per condition

3. Results

Environmental feedback changed trust in opposite directions under different conditions (Fig. 2). We compared branches that continued Q -learning from the same learned state at

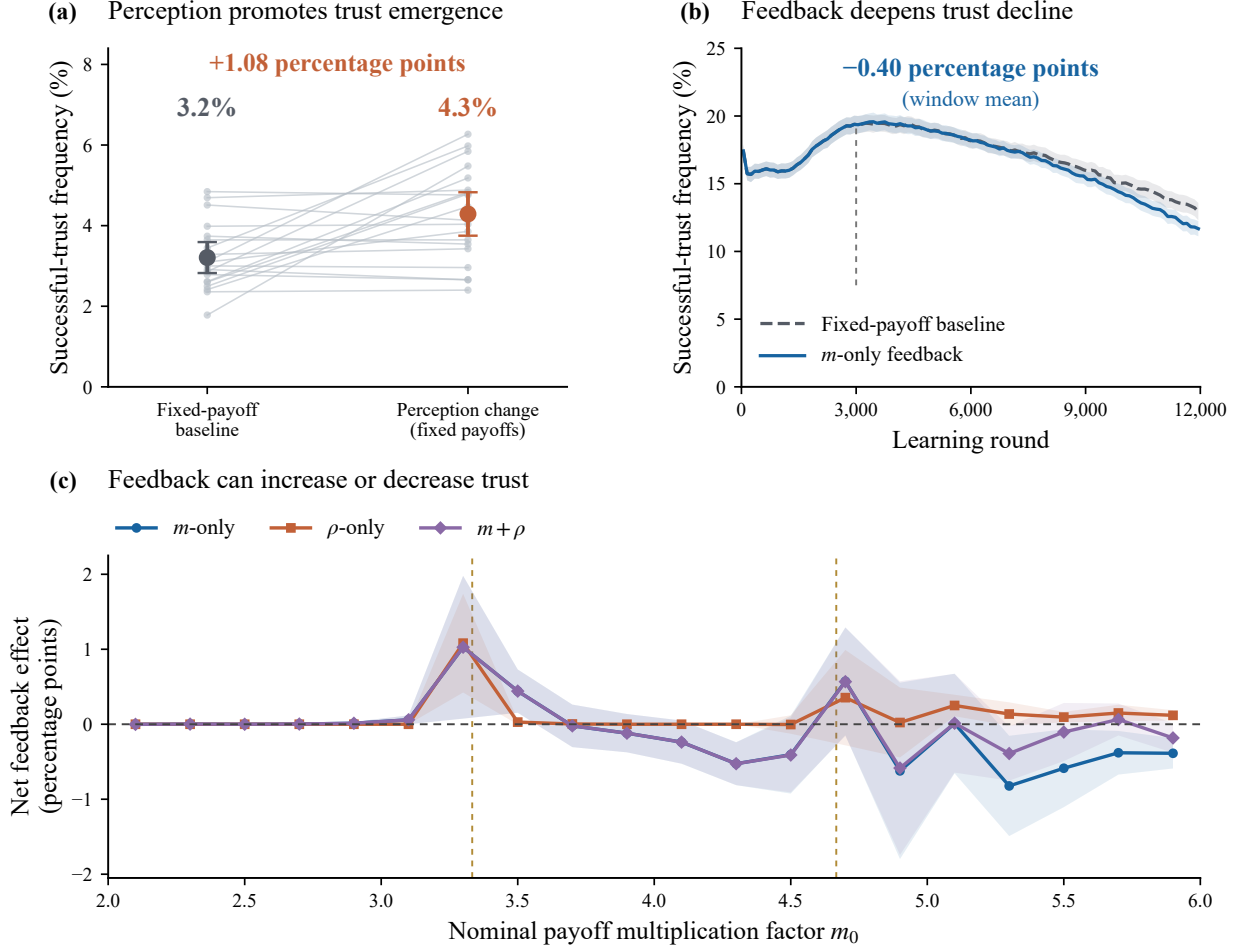


Figure 2: Environmental feedback promotes trust emergence or deepens trust decline under different conditions. All panels use $\rho_0 = 0.2075$. Net feedback effects are within-seed differences in mean successful-trust frequency from the fixed-payoff baseline. (a) At $m_0 = 3.3$, the ρ -only branch changes the perceived environmental state while both realized payoff parameters remain fixed. Thin lines connect paired outcomes; large points and error bars show means and 95% Student-*t* intervals. The paired increase is 1.08 percentage points (95% interval: 0.42–1.74). (b) Baseline and *m*-only trajectories at D ($m_0 = 5.7$), shown as 100-round means. The vertical dashed line marks feedback activation at round 3,000. The analysis-window paired change is -0.40 percentage points (95% interval: -0.56 to -0.24). (c) Net feedback effects across nominal payoff multiplication factors. The *m*-only, ρ -only and joint branches allow *m*, ρ , or both payoff parameters to respond to each branch’s environment, which also determines the perceived state. The horizontal dashed line denotes zero effect; gold vertical lines mark the perception-tier boundaries. Panels (a,c) use 20 paired seeds; (b) uses 50 additional paired seeds. Shading in (b,c) represents pointwise 95% Student-*t* intervals. Window means use rounds 3,000–11,999. Perception promotes trust emergence at fixed payoff parameters in (a), whereas feedback deepens the decline in (b).

round 3,000; the static baseline kept the payoff parameters and environmental state fixed. We measured trust by the successful-trust frequency f_{11} , the share of interactions in which the investor invested and the trustee acted trustworthily. The net feedback effect was

$$\Delta \bar{f}_{11} = \bar{f}_{11}^{\text{feedback}} - \bar{f}_{11}^{\text{base}}.$$

The bars denote means over rounds 3,000–11,999 and paired seeds. Positive and negative effects indicate higher and lower trust than the baseline, respectively.

At $\rho_0 = 0.2075$ and $m_0 = 3.3$, changing the perceived environmental state in the ρ -only branch increased successful-trust frequency from 3.2% to 4.3%, a paired increase of 1.08 percentage points (Fig. 2a). Both realized payoff parameters remained fixed: perception alone promoted the emergence of trust. At point D ($m_0 = 5.7$), m -only feedback instead deepened the decline in trust, reducing the analysis-window mean by 0.40 percentage points relative to the baseline (Fig. 2b). Across the three feedback branches, the direction and magnitude of the response varied with the nominal payoff multiplication factor (Fig. 2c). We next examine how the payoff and perception channels, state representation, and learning history produce these responses.

3.1. Baseline finite-horizon crossover

Over the 12,000-round horizon, baseline successful trust increased with m from a low-trust to a higher-trust range (Fig. 3a,b). Logistic fits to the ten sampled ρ slices gave $R^2 = 0.990$ – 0.997 , with fitted midpoints $m^* = 4.22$ – 5.13 and widths $w = 0.67$ – 0.99 . Midpoints generally shifted toward larger m as the defect ratio increased, with departures from monotonicity between adjacent slices. The fitted band defines the three ranges in Fig. 3d. Successful trust remained a minority outcome even in the higher-trust range ($\bar{f}_{11} \leq 0.21$), providing the baseline scale for the feedback comparisons.

3.2. Baseline learning drift across the crossover

Baseline successful-trust trajectories continued to evolve after the burn-in cutoff (Fig. 4). At $m = 3.2$, the post-cutoff block mean remained near an exploration-dominated low-trust level, changing from 0.025 at the cutoff to 0.028 in the final block. Across $m = 3.8$ – 5.6 , ensemble means peaked between rounds 3,150 and 3,980 at 0.086–0.194, then declined to 0.040–0.131 in the final block. The baseline learning drift δ defined in Section 2.4 was $+0.001 \pm 0.001$ at $m = 3.2$ and -0.017 to -0.020 at $m = 3.8$ – 5.6 . All intervals in the latter range excluded zero. The reference window [2000, 3000) lay near the peaks of these ensemble means, which remained below their reference averages for much of the analysis window. Equation (11) uses each seed’s deviation from its own reference; the plotted curves show ensemble means.

3.3. Closing the loop: two channels in the response landscape

Feedback produced small, parameter-dependent changes in successful trust (Fig. 5). Across the 400 grid points, the paired m -only change in $\bar{f}_{11}$ ranged from -0.015 to $+0.017$ with a median absolute value of 0.002; the pointwise intervals lay above zero at 49 points, below zero at 96, and included zero at 255. At the sampled $m \leq 3.1$ points, the population remained at an exploration-dominated low-trust level. The maximum absolute paired mean in this range was approximately 0.0021. Two narrow regions of positive mean response extended across sampled ρ values at $m = 3.3$ and $m = 4.7$. These columns adjoin the perception-tier edges at 3.333 and 4.667. They contained 21 of the 49 intervals above zero and none of the 96 intervals below zero. Including neighboring columns at 3.5 and 4.9

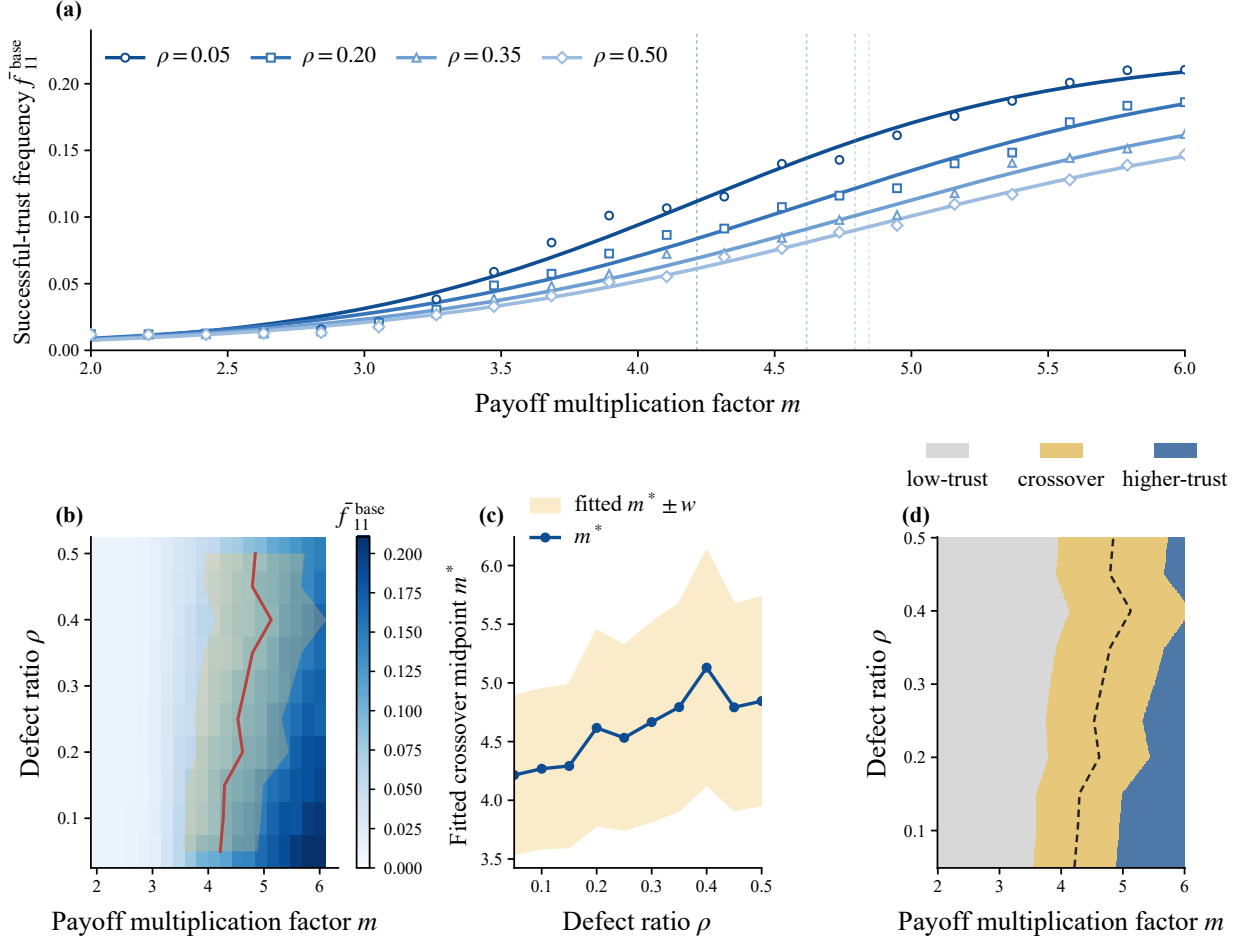


Figure 3: Baseline behavior over the fixed 12,000-round horizon. (a) Baseline grid means $\bar{f}_{11}(m)$ at selected ρ (markers) and four-parameter logistic fits (lines). (b) Finite-horizon map $\bar{f}_{11}^{\text{base}}(m, \rho)$ with fitted midpoints and the $m^* \pm w$ band. (c) Fitted midpoint $m^*(\rho)$ with shading spanning $m^* \pm w$. (d) Fitted crossover band at $T = 12,000$, separating the low-trust, crossover, and higher-trust ranges. Each grid value is the mean of 20 seeds over rounds 3,000–11,999.

increased the first count to 31; at $m = 3.3$, 16 of 20 intervals lay above zero. Over the interior columns $m \geq 3.7$, excluding these boundary columns and their neighbors, the mean change was -0.003 . Of these 200 cells, 93 intervals lay below zero and 11 above.

We fixed $\rho_0 = 0.2075$ and compared successful trust across the three feedback branches (Fig. 5c and Table 2). At $m_0 = 3.3$, near the lower perception-tier boundary, the m -only change was $+0.0103 \pm 0.0095$, close to the fixed-payoff perception response shown in Fig. 2a. The m -only mean excursion was $+0.060$, with temporal standard deviation 0.052 and nominal distance 0.033 to the lower tier boundary.

At $m_0 = 4.7$, 0.033 above the second edge, both branch point estimates were positive: $+0.0035$ for ρ -only and $+0.0057$ for m -only. Both 95% intervals spanned zero at this operating point. At $m_0 = 4.5$, 0.167 below that edge, the mean excursion was directed away from it ($\Delta \bar{m} = -0.198$). The ρ -only mean was close to zero and the m -only mean was negative; both 95% intervals spanned zero. For $m_0 \leq 4.1$, the m -only and $m+\rho$ full-window successful-trust means agreed to numerical precision within each seed, consistent with the centered ρ -tier remaining nominal.

At the listed $m_0 \geq 4.9$ points, ρ -only point estimates were small and positive ($+0.0002$ to $+0.0015$), opposite to the m -only estimates. These branches allow a declining environ-

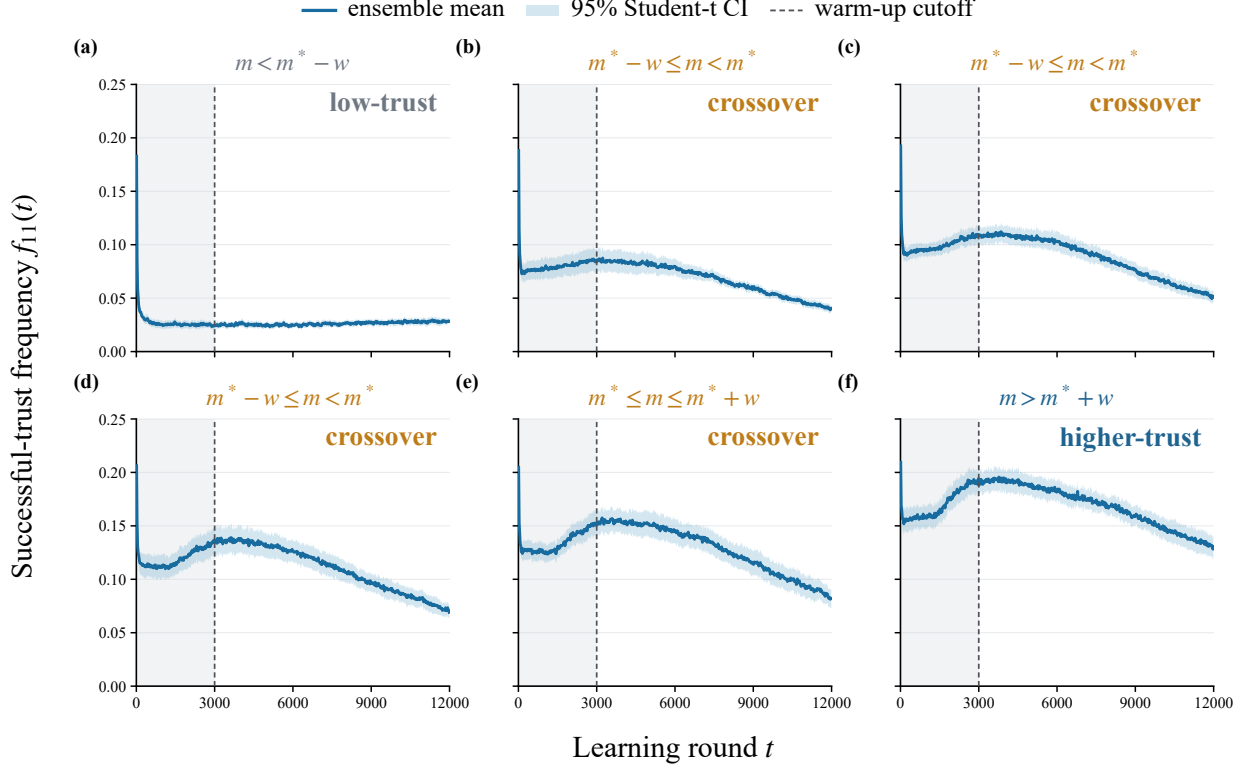


Figure 4: Baseline successful-trust trajectories across six payoff multiplication factors at $\rho = 0.2075$. Panels (a–f) show $f_{11}(t)$ at $m = 3.20, 3.80, 4.20, 4.60, 5.00$, and 5.60 , respectively, on common axes. Regime labels use the fitted midpoint $m^* = 4.62$ and width $w = 0.83$. The normalized positions $(m - m^*)/w$ in (a–f) are approximately $-1.71, -0.99, -0.50, -0.02, +0.46$, and $+1.19$. Within-seed 20-round block averages are summarized by the ensemble mean and two-sided 95% Student- t confidence interval across 20 independent seeds. Shading and the dashed line mark the specified burn-in interval and cutoff at round 3,000. For $m \geq 3.8$, ensemble means peak within about 1,000 rounds of the cutoff and then decline over the remaining $T = 12,000$ horizon.

Table 2: Channel-control comparisons along the fixed $\rho_0 = 0.2075$ slice (20 matched seeds; means $\pm$ 95% Student- t half-widths for the m -only and ρ -only paired changes $\Delta \bar{f}_{11}$). The joint-channel column reports means; its intervals are shown in Fig. 5c. d_{edge} is the distance from the nominal m_0 to the nearest perception-tier edge; $\Delta \bar{m}$ and s_m are the window mean excursion and temporal standard deviation of the realized $m(t)$ in the m -only branch. The ρ -only branch realizes m_0 in every round; its ρ -tier stays in the middle band for $m_0 \leq 4.1$, where the m -only and $m+\rho$ full-window successful-trust means agree to numerical precision within each seed.

m_0	d_{edge}	$\Delta \bar{f}_{11}$ (m -only)	$\Delta \bar{f}_{11}$ (ρ -only)	$\Delta \bar{f}_{11}$ ($m+\rho$)	$\Delta \bar{m}$	s_m
3.1	0.233	$+0.0006 \pm 0.0006$	0	$+0.0006$	$+0.006$	0.015
3.3	0.033	$+0.0103 \pm 0.0095$	$+0.0108 \pm 0.0066$	$+0.0103$	$+0.060$	0.052
3.5	0.167	$+0.0044 \pm 0.0029$	$+0.0003 \pm 0.0003$	$+0.0044$	$+0.013$	0.057
4.1	0.567	-0.0024 ± 0.0029	0	-0.0024	-0.131	0.175
4.5	0.167	-0.0041 ± 0.0052	-0.00004 ± 0.0012	-0.0041	-0.198	0.212
4.7	0.033	$+0.0057 \pm 0.0072$	$+0.0035 \pm 0.0064$	$+0.0057$	-0.054	0.117
4.9	0.233	-0.0062 ± 0.0118	$+0.0002 \pm 0.0047$	-0.0058	-0.221	0.229
5.3	0.633	-0.0082 ± 0.0067	$+0.0014 \pm 0.0016$	-0.0039	-0.216	0.234
5.7	1.033	-0.0038 ± 0.0029	$+0.0015 \pm 0.0011$	$+0.0006$	-0.130	0.139

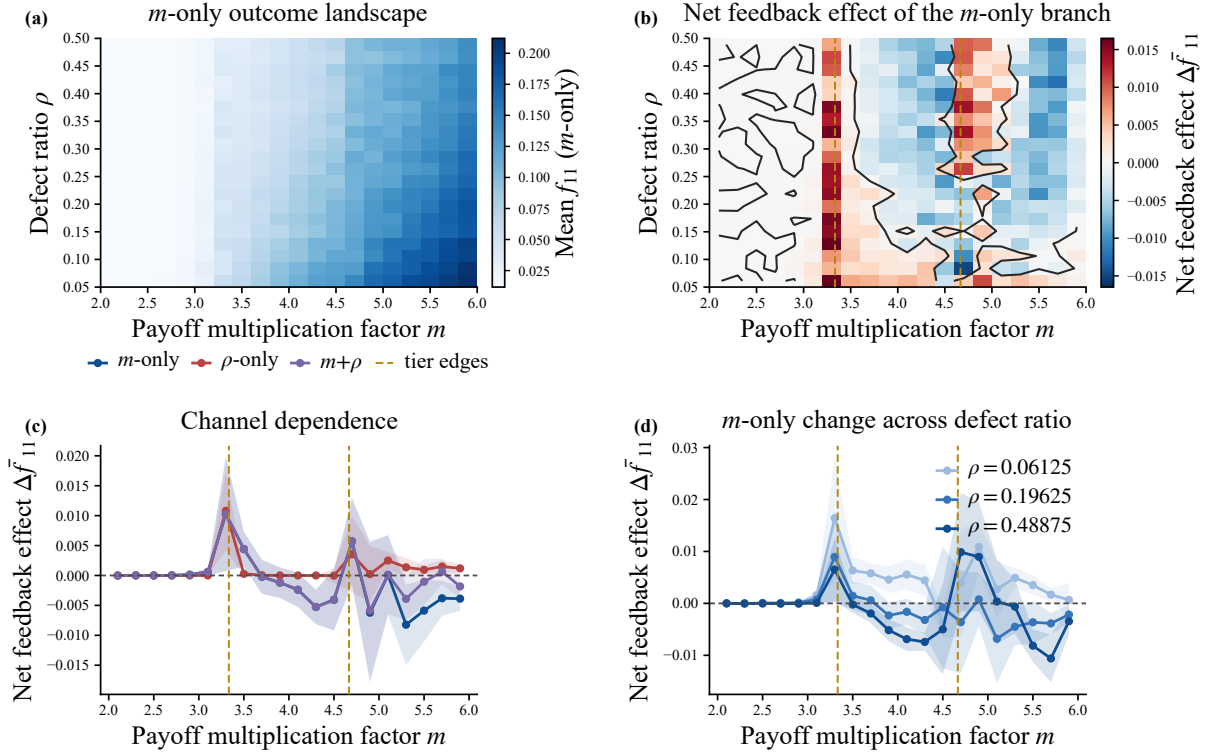


Figure 5: Paired response landscape and payoff-channel controls. (a) Mean successful trust frequency $\bar{f}_{11}$ for the m -only branch on the 20×20 (m, ρ) grid. (b) Within-seed m -only-minus-baseline change on the same grid; black contours mark zero mean change. Dashed gold lines in (b–d) mark the perception-tier edges at $m = 3.333$ and 4.667 . (c) m -only, ρ -only, and joint-channel changes along the fixed $\rho_0 = 0.2075$ slice. (d) m -only changes along three sampled ρ profiles. Grid values in (a,b) are means over 20 paired seeds; lines and bands in (c,d) show means and pointwise 95% Student- t confidence intervals across the same 20 matched seeds.

ment to lower the defect-ratio tier. Away from the highlighted boundary columns, declining mean payoff multiplication factors accompanied predominantly negative m -only response estimates, consistent with a contribution from payoff feedback. The mean decline in the payoff multiplication factor was 0.13–0.22 at the listed non-boundary points from $m_0 = 4.1$ to 5.7.

3.4. Baseline learning drift and environmental response at four parameter settings

We examined four parameter settings using 50 additional random seeds, separate from those used for the parameter grid. Figure 6 shows the baseline learning drift and environmental response at these settings. At A ($m = 3.1$), baseline learning drift was -0.0005 ± 0.0006 , the mean environmental displacement was $\langle E - E_0 \rangle = +0.001$, and the paired change was $+0.0003$ (95% CI $[-0.0002, +0.0007]$). At B, C, and D, baseline learning drift was -0.021 , -0.021 , and -0.018 , respectively, with mean environmental declines of 0.041, 0.058, and 0.076. The realized payoff multiplication factor declined by 0.16, 0.21, and 0.12 on average over the window, and by 0.41, 0.44, and 0.40 in the final bin. The net feedback effects were -0.0045 $[-0.0067, -0.0024]$, -0.0031 $[-0.0084, +0.0022]$, and -0.0040 $[-0.0056, -0.0024]$, respectively (95% CIs; Table D.5).

Environmental excursions were positively associated with baseline learning drift across seeds. Across the 50 seeds, the Pearson coefficients were 0.76, 0.87, 0.76, and 0.87 at A–D, respectively. The correlations between realized excursions $\Delta \bar{m}$ and net feedback effects

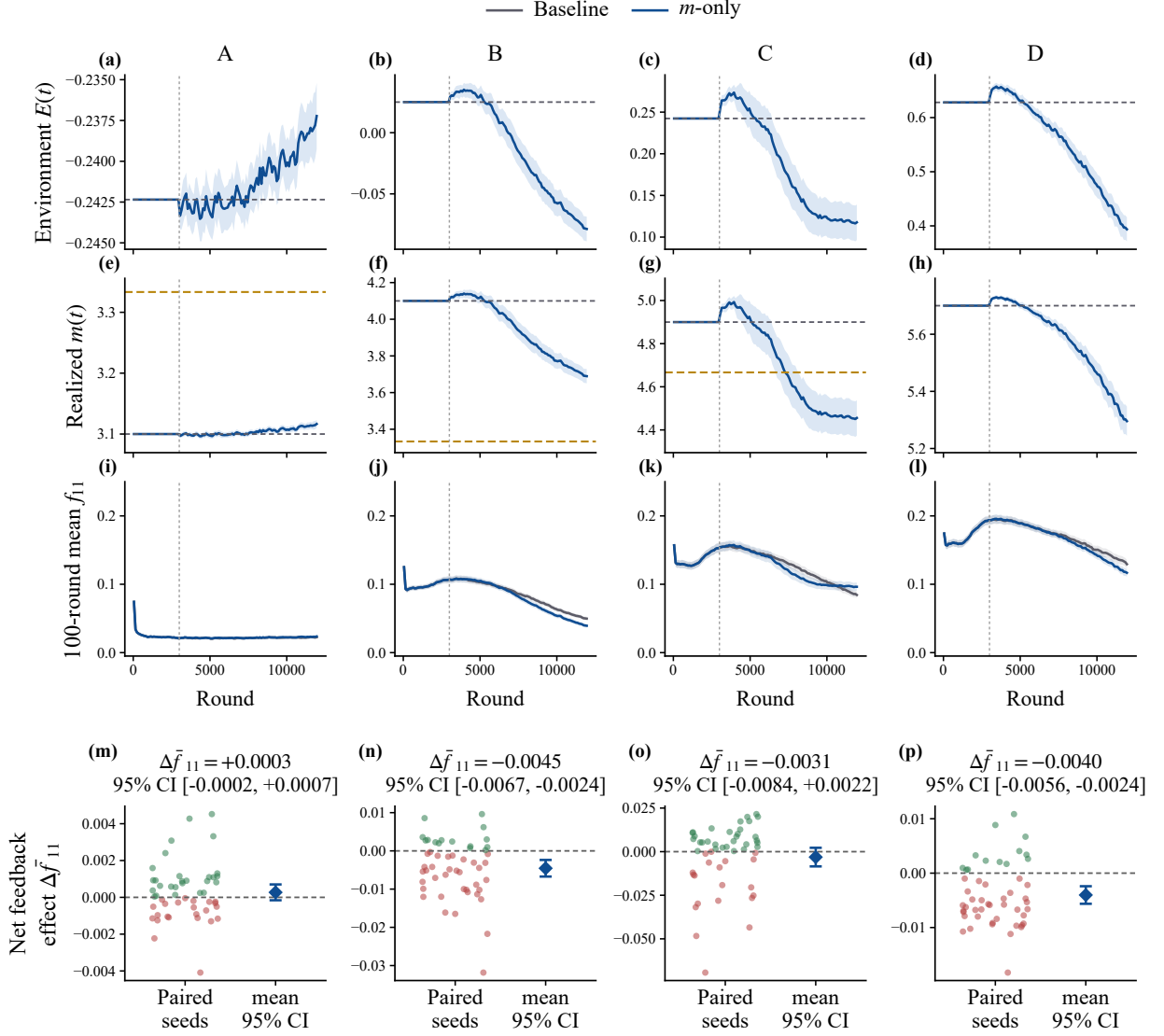


Figure 6: Parameter settings A–D (columns) at $\rho_0 = 0.2075$ and $m_0 = 3.1, 4.1, 4.9$, and 5.7 . (a–d) Environment $E(t)$ of the m -only branch with the nominal E_0 (dashed grey). (e–h) Realized $m(t)$ with the nominal m_0 (dashed grey) and the perception-tier edges (dashed gold), summarized as ensemble means. (i–l) Baseline and m -only f_{11} trajectories. (m–p) Within-seed m -only-minus-baseline analysis-window change for the 50 seeds (green positive, red negative) with the mean and 95% Student- t interval. Curves and bands in (a–l) are 100-round means and pointwise 95% Student- t intervals across 50 matched seeds (100–149); the vertical dotted line marks the feedback start at round 3000 and the fixed analysis window is [3000, 12000).

were 0.65, 0.91, 0.94, and 0.66. These are across-seed associations relative to the shared pre-feedback reference.

At C, inside the fitted crossover band, the excursion–effect correlation was strongest, while the interval for the mean feedback change included zero. The three seeds whose environments rose had paired changes from $+0.013$ to $+0.022$; the seed whose realized m fell by 1.1 changed by -0.069 . All twelve seeds with $\Delta \bar{m} < -0.3$ changed negatively, while 60% of the seeds changed positively. At C, the same nominal operating point therefore produced feedback responses of both signs across seeds.

4. Mechanisms and robustness

4.1. Microscopic reorganization across the crossover

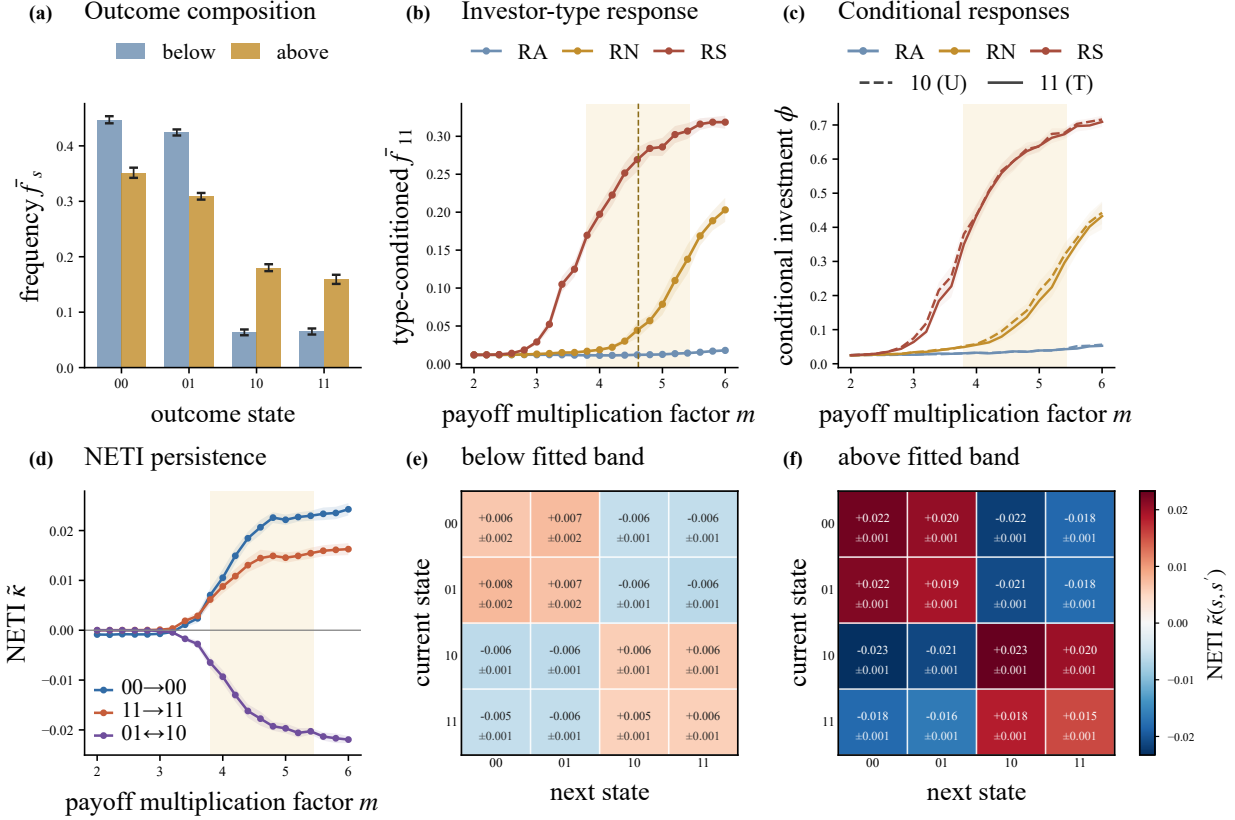


Figure 7: Finite-horizon mechanism atlas at $\rho = 0.2075$. (a) Outcome-state composition below ($m^* - w = 3.789$) and above ($m^* + w = 5.444$) the nearest fitted crossover. (b) Type-conditioned successful trust across m . (c) Type-resolved investment conditional on the previous outcome: dashed lines denote outcome 10 (untrustworthy, U), and solid lines denote outcome 11 (trustworthy, T). In (b,c), RA, RN, and RS denote risk-averse, risk-neutral, and risk-seeking investors. (d) Three NETI signatures across m : $00 \rightarrow 00$, $11 \rightarrow 11$, and the mean $01 \leftrightarrow 10$ cross-block entry. (e,f) Full NETI matrices below and above the fitted band, with one shared color scale and cellwise confidence intervals. Curves, bars, and ribbons are means and two-sided 95% Student- t confidence intervals across 20 independent static-baseline seeds over rounds 3,000–11,999 of the fixed $T = 12,000$ horizon. The gold region denotes the fitted $m^* \pm w$ band.

We examined investor participation and outcome composition across the fitted crossover. Across this band, the non-investment outcomes 00 and 01 decreased from 0.447 ± 0.006 and 0.424 ± 0.005 to 0.351 ± 0.009 and 0.309 ± 0.006 , whereas 10 and 11 increased from 0.064 ± 0.005 and 0.065 ± 0.005 to 0.180 ± 0.006 and 0.159 ± 0.008 (Fig. 7a). The type-resolved change was concentrated in RN and RS investors. RN $\bar{f}_{11}$ increased from 0.017 ± 0.001 to 0.148 ± 0.015 and RS $\bar{f}_{11}$ from 0.165 ± 0.012 to 0.312 ± 0.008 ; RA remained low (0.011 ± 0.000 to 0.015 ± 0.001). Conditional investment frequencies after outcomes 10 and 11, respectively, increased from (0.050, 0.048) to (0.333, 0.306) for RN and from (0.366, 0.328) to (0.689, 0.677) for RS, while the corresponding RA values changed only from (0.031, 0.031) to (0.046, 0.044) (Fig. 7b,c). Values are means $\pm$ 95% Student- t confidence interval half-widths; the smallest per-seed conditional denominator was 6,247. RS investors lead the response, with an increasing RN contribution at larger m .

4.2. Pooled outcome-pair structure: the NETI view

We compared NETI across values of the payoff multiplication factor m and found stronger excess self-pairing for both 00 and 11 across the crossover. At $m = 2.0$, $\tilde{\kappa}(00, 00) = -0.0009 \pm 0.0001$ and $\tilde{\kappa}(11, 11)$ was approximately zero. By $m = 4.6$, the two entries had increased to 0.021 ± 0.001 and 0.014 ± 0.002 , respectively, while the mean $01 \leftrightarrow 10$ entry declined to -0.018 ± 0.001 . At $m = 6.0$, the corresponding values were 0.024 ± 0.001 , 0.016 ± 0.001 , and -0.022 ± 0.001 . The matrices below and above the fitted crossover band showed the same pattern: $\tilde{\kappa}(00, 00)$ increased from 0.006 ± 0.002 to 0.022 ± 0.001 , and $\tilde{\kappa}(11, 11)$ from 0.006 ± 0.001 to 0.015 ± 0.001 (Fig. 7d–f). These NETI changes describe pooled population-level pair frequencies. Values are means $\pm$ 95% Student- t confidence interval half-widths over 20 seeds.

4.3. Myopic thresholds and type-resolved participation

We define $\hat{q}$ as the perceived trustworthiness probability and derive a myopic benchmark from immediate subjective payoffs. For type θ , investing has subjective value $\hat{q}(m/2 - 1) - (1 - \hat{q})\lambda_\theta$ for $m \geq 2$, giving the risk-adjusted trust threshold

$$\tilde{q}_\theta^*(m) = \frac{2\lambda_\theta}{m + 2\lambda_\theta - 2}, \quad \tilde{q}_{\text{RA}}^* > \tilde{q}_{\text{RN}}^* > \tilde{q}_{\text{RS}}^* \quad (m > 2). \quad (15)$$

At $m = 2$ every type shares the threshold $\tilde{q}_\theta^* = 1$. We use this static benchmark to predict the participation ordering (Appendix A.2).

For a trustee receiving positive inflow $I_j > 0$, the untrustworthy action yields an immediate monetary advantage $\rho m I_j / 2$. The trustworthy action instead yields an additional reputation increment η_R , independent of the current reputation (Appendix A.2). Trustworthiness is therefore weakly preferred in the myopic comparison when

$$\lambda_\phi \geq \lambda_\phi^* = \frac{\rho m I_j}{2\eta_R}. \quad (16)$$

At zero inflow, the two actions are myopically indifferent. On the fixed- $\rho_0 = 0.2075$ slice, the crossing $m_{\text{cross}} = 2\eta_R\lambda_\phi/(\rho_0 I_j)$ lies below 2 for both trustee types in Table 1, even at $I_j = 1$. This predicts no trustee-type onset within $m \in [2, 6]$ on that slice. Targeted checks with $\eta_R = 0.2$ or $\lambda_L = 5$ also failed to produce a theory-aligned L–S trustworthiness response.

At $m = 2$, all three investor types remained near a common exploration-dominated floor ($\bar{f}_{11} \simeq 0.012$). By $m = 4.6$, type-conditioned $\bar{f}_{11}$ was 0.012 ± 0.001 , 0.044 ± 0.011 , and 0.269 ± 0.016 for RA, RN, and RS, respectively (mean $\pm$ 95% confidence-interval half-width). At the same m , investment after an untrustworthy outcome (10) was 0.036 ± 0.002 , 0.126 ± 0.023 , and 0.595 ± 0.025 , and investment after a successful outcome (11) was 0.036 ± 0.002 , 0.107 ± 0.020 , and 0.596 ± 0.030 . RS led the rise through most of the sampled range, RN followed at larger m , and RA remained low. This ordering emerged above the shared exploration floor (Fig. 7b,c).

4.4. Continuation value advances the risk-seeking onset

We measured conditional trustworthiness q_0 in the low- m region, away from the fitted crossover. Inverting Eq. (15) at this value gives the static onset benchmark

$$m_1^*(\lambda, q_0) = 2 + \frac{2\lambda(1 - q_0)}{q_0}, \quad (17)$$

which we evaluate for the risk-seeking type at $m_1^*(\lambda_{\text{RS}}, q_0)$. Across the nine (γ, ε) settings, the independently measured $q_0 = 0.486\text{--}0.502$ gave benchmark onsets $m_{\text{theory}}^* = 2.99\text{--}3.06$,

whereas logistic fits to the pooled three-type response gave midpoints $m_{\text{fit}}^* = 4.25\text{--}4.70$. The offset $m_{\text{fit}}^* - m_{\text{theory}}^*$ was therefore positive in every cell (+1.26 to +1.65), with all 95% whole-seed bootstrap intervals excluding zero (Fig. 8a).

Increasing λ_{RS} from 0.30 to 0.80 raised both the pooled fitted midpoint ($3.93 \rightarrow 5.37$) and the RS benchmark ($2.56 \rightarrow 3.69$). Their offset remained +1.37 to +1.68, with all bootstrap intervals above zero (Fig. 8b). The pooled midpoint and the RS one-round reference thus shifted in the same direction as loss aversion increased.

We tested the contribution of investor continuation value by setting $\gamma_{\text{inv}} = 0$ while retaining Q -table updates (Appendix E.3, Fig. S6). We compared both RS fitted midpoints with a common one-round reference estimated at $\gamma_{\text{inv}} = 0.9$. The zero-discount midpoint was 4.24 ± 0.08 , a $+1.21 \pm 0.08$ gap above this reference, $m_1^* = 3.035 \pm 0.031$. Restoring continuation value lowered the midpoint to 3.61 ± 0.04 , a paired shift of -0.63 ± 0.08 (estimates $\pm$ SE). Continuation value therefore advanced the RS crossover.

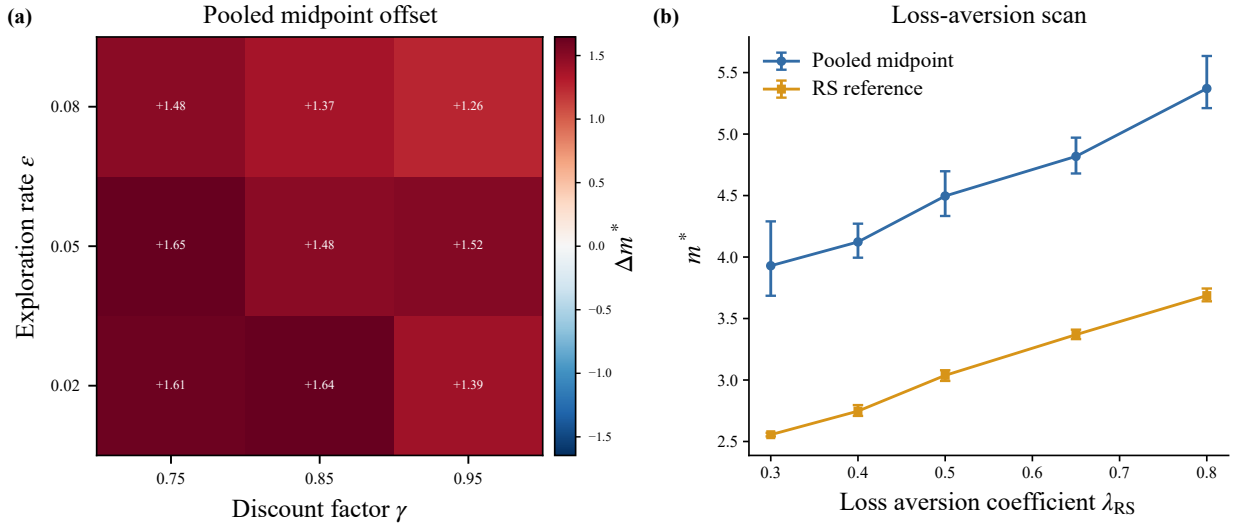


Figure 8: Pooled fitted midpoints and the RS one-round reference. (a) Offset $\Delta m^* = m_{\text{fit}}^* - m_{\text{theory}}^*$ across the (γ, ϵ) plane. (b) Pooled midpoints and RS references across the different loss-aversion coefficients λ_{RS} . Error bars are two-sided 95% whole-seed bootstrap confidence intervals from 1,000 resamples. For every setting, $q_0 = N_{11}/(N_{10} + N_{11})$ is measured independently at fixed $m = 3.0$ from 20 seeds over rounds 3,000–11,999; all panels use the static baseline at $\rho = 0.2075$. The fitted midpoint summarizes the three-type population mixture, whereas the reference is the RS one-round utility zero. Both increase with the RS loss aversion coefficient.

4.5. A quasi-static description of the payoff response

Centering sets the environmental drive to zero at the reference observables. We construct a quasi-static scalar approximation from smooth static-response maps within unchanged payoff and observation tiers (Appendix A.3). Its dimensionless loop gain is

$$G_{\mathcal{W}}(m_0, \rho_0) = M'(E_0) [\beta_C \chi_{C,m}^{\mathcal{W}} + \beta_r \chi_{r,m}^{\mathcal{W}}], \quad (18)$$

where $M'(E_0) = k(m_{\text{max}} - m_{\text{min}})z_0(1 - z_0)$ is the sigmoid slope. The susceptibilities $\chi_{C,m}^{\mathcal{W}}$ and $\chi_{r,m}^{\mathcal{W}}$ are derivatives of the smooth approximations to finite-window successful trust and mean investor payoff. For a stable scalar map, a sustained forcing is amplified by $1/(1 - G_{\mathcal{W}})$. Its discrete multiplier is $a = 1 - \nu(1 - G_{\mathcal{W}})$; for $0 < a < 1$, the relaxation time is $\tau = -1/\log a \simeq 1/[\nu(1 - G_{\mathcal{W}})]$.

We applied a small environmental impulse in a checkpoint-fork control (Appendix E.1, Fig. S4). At the mid-band operating point, the response decayed with a fitted $\tau = 41.45$ rounds (SE 3.51). The discrete-time conversion gave $G_{\text{eff}} = 0.205$ (approximate 95% CI [0.020, 0.332]). In the scalar model, this positive gain slows the decay of environmental perturbations.

The m -only change in successful trust agreed in sign with the static-baseline benchmark at 10 of 11 selected nominal points (Appendix A.3). We obtained the benchmark change by multiplying the baseline finite-difference slope by the realized mean excursion $\Delta\bar{m}$. The median observed-to-benchmark ratio was 0.42 (interquartile range [0.252, 0.642]), so the observed changes were typically smaller in magnitude than the finite-grid benchmark.

4.6. Perception responses near tier boundaries

The largest positive slice responses occurred near $m_0 = 3.3$ and 4.7, each only 0.033 from a tier edge, compared with temporal standard deviations of 0.05–0.12 in realized $m(t)$ (Table 2). At $m_0 = 3.3$, the ρ -only branch changed perception while both payoff parameters stayed fixed. Its positive response was comparable in size to the m -only response. At 4.7, both payoff parameters also stayed nominal, but the positive point estimate had an interval spanning zero. At 4.5, the nearest edge was five times farther away and the mean excursion was directed away from it; both branch intervals spanned zero.

First entry into a tier absent from the fixed-environment burn-in activates unupdated Q rows. Uniform tie-breaking gives each binary action probability 1/2 on its first selection, which can initially raise investment above a low learned rate. The full-window contrast combines first visits, revisits, and subsequent learning.

We separated payoff and perception effects in a 2×2 factorial experiment (Appendix E.2, Fig. S5). Perception responses were nonzero in individual runs near tier boundaries, whereas dynamic and fixed perception produced identical successful-trust trajectories at the tested mid-band point. The interaction interval spanned zero and effects of either sign comparable in magnitude to the main effects. At $m_0 = 3.333$, the perception response was nonzero under $\kappa = (1/3, 2/3)$ and exactly zero under $\kappa = (0.25, 0.75)$ (Appendix E.4, Fig. S7). These results identify the observed environmental tier as a route to behavioral change. The between-scheme comparison combines changes in state discretization, the ρ -switching rule and the resulting learning histories.

4.7. Feedback preserves the pooled two-block sign pattern at B–D

Under feedback, the pooled NETI matrices retained their principal two-block sign pattern at B–D, while A remained weakly structured near the low-trust level (Fig. 9). Across the 64 paired NETI cells, mean differences ranged from -0.0024 to $+0.0024$; 21 pointwise intervals lay above zero, 20 below zero, and 23 included zero. The largest paired difference at each point was 4%, 6%, 11%, and 2% of the largest baseline entry at A–D.

NETI retained the two-block sign pattern at B–D under conditional independence benchmarks formed from population outcome margins within 100-round blocks or each adjacent pair of rounds (Appendix A.1). A remained weakly structured near its low-trust level.

4.8. Robustness of the finite-horizon feedback response

With $\beta_C = \beta_r = 0$, we obtained the same successful-trust trajectory as in the static baseline (Figs. 10 and 11). Along the diagonal of increasing drive weights $(\beta_C, \beta_r) = (0.7, 0.6)$, $(1.4, 1.2)$, $(2.1, 1.8)$, the net feedback effect at D was -0.0006 , -0.0049 , and -0.011 ; at A, on the floor, it was $+0.0008$, $+0.0007$, and $+0.0014$ (Fig. 11a,b). Across the 16 cells, five

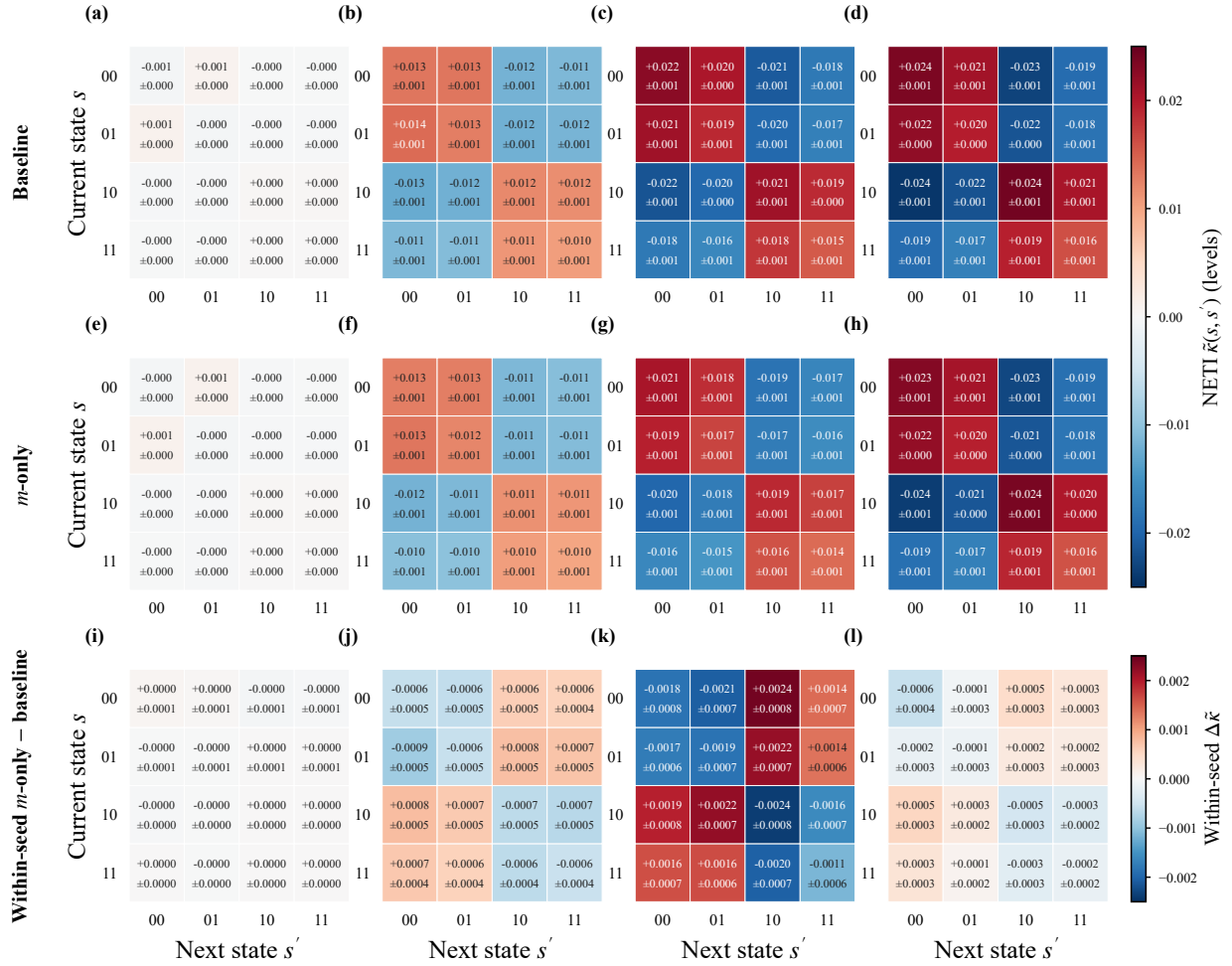


Figure 9: Finite-window NETI matrix comparison at the four parameter settings. Panels (a–d), (e–h), and (i–l) form the baseline, m -only, and within-seed m -only-minus-baseline rows, respectively; columns follow A–D, with $m_0 = 3.1, 4.1, 4.9$, and 5.7 , respectively, at $\rho_0 = 0.2075$. The first two rows share a symmetric level scale and the contrast row has its own symmetric difference scale, ten times finer. Cell text reports the mean and two-sided 95% Student- t confidence interval half-width across 50 matched seeds over $[3000, 12000)$. NETI reports finite-window, cellwise joint-frequency excess relative to a shuffled benchmark.

pointwise intervals at D lay below zero and five at A above zero; the interval at the main drive weights included zero at A and lay below zero at D.

At D, extending the horizon made the net feedback effect more negative, from -0.0049 at 12k to -0.0063 at 24k and -0.012 at 36k rounds. All three intervals were below zero. Restricting the observation window to $[6000, 12000)$ changed the effect from -0.0049 to -0.0068 (Fig. 10c,f). At A the 12k and 36k intervals included zero and the 24k mean was $+0.0012$. The payoff response thus continued to accumulate over the tested horizons at D, where the baseline learning decline also persisted.

Extending the reference window from $[2000, 3000)$ to $[1000, 3000)$ includes the rising part of the trajectory and therefore lowers the reference average. With this longer reference window, the net feedback effect at D changed from -0.0049 (95% CI -0.0077 to -0.0020) to -0.0003 (-0.0029 to $+0.0024$; Fig. 11h), attenuating the estimated response toward zero.

Varying the environmental update rate ν over 0.01 , 0.03 , and 0.07 gave net feedback effects of -0.0039 , -0.0049 , and -0.0021 at D, respectively (Fig. 11d). Adding drive noise $\sigma_\xi = 0.002$ or 0.005 gave -0.0021 and -0.0032 at D, respectively (Fig. 11f). At A, the effects remained between $+0.0007$ and $+0.0010$ under both variations (Fig. 11c,e). The estimated net

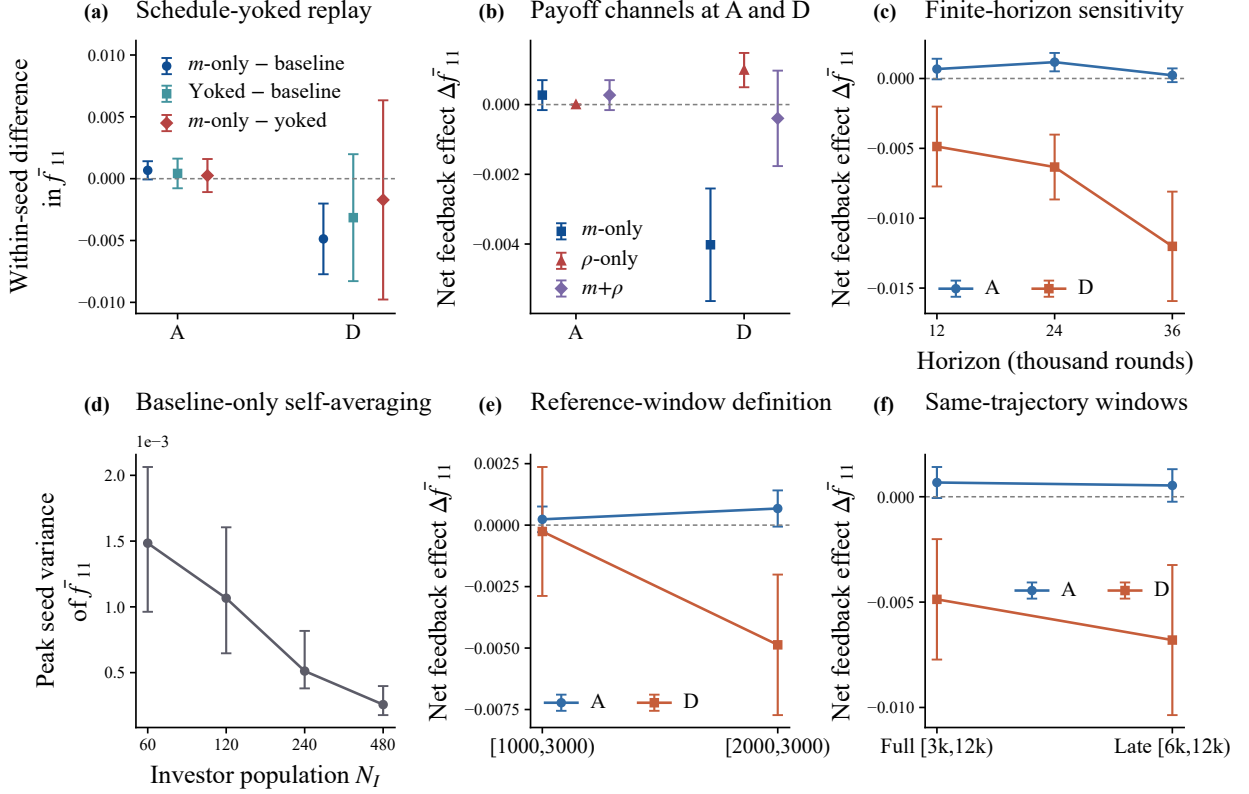


Figure 10: Controls on the centered finite-horizon response at points A and D. (a) Within-seed m -only-minus-baseline, foreign-schedule-yoked-minus-baseline, and m -only-minus-yoked contrasts. (b) Within-seed changes of the m -only, ρ -only, and joint-channel branches relative to the baseline; the ρ -only change at D is positive while the m -only change is negative. (c) Paired m -only changes at 12k, 24k, and 36k rounds, each summarized over $[3000, T)$. (d) Baseline-only peak seed variance across four population sizes. (e) Net feedback effects under reference windows $[1000, 3000)$ and $[2000, 3000)$. (f) Full $[3000, 12000)$ and fixed late $[6000, 12000)$ observation windows using the reference window $[2000, 3000)$. Panels (a,c,e,f) use 20 matched seeds. Intervals involving yoked replay in (a) use the alternating-group Bonferroni procedure in Appendix C (approximate pointwise 95% coverage). The other intervals in (a,c,e,f) are t_{19} intervals; panel (b) uses 50 matched seeds and t_{49} intervals. Panel (d) uses 20 baseline seeds and 1,000 whole-seed bootstrap replicates. All intervals are pointwise.

feedback effects at D remained negative across these settings, with some confidence intervals including zero.

At D the mean m -only, ρ -only, and joint changes relative to the baseline were -0.0040 , $+0.0010$, and -0.0004 (Fig. 10b). The ρ -only point estimate has the opposite sign to the m -only estimate, and the joint estimate lies between them. For these branches, over $[3000, 12000)$ and the 50 matched seeds, the paired departure from additivity $\Delta \bar{f}_{11}^{m+\rho} - (\Delta \bar{f}_{11}^m + \Delta \bar{f}_{11}^\rho)$ is $+0.0026$ (95% CI $[+0.0018, +0.0034]$). Each branch evolves its own environment and perceived state; the factorial control in Appendix E.2 instead switches payoff and perception separately. The combined branch response at D differed from the sum of the two separate responses over this window.

We used schedule-yoked replay to apply each seed’s environmental schedule to the preceding seed’s learners, forming a cycle of 20 recipient–donor pairs. Relative to baseline, the m -only contrasts were $+0.0007$ at A (95% CI $[-0.0001, +0.0014]$) and -0.0049 at D ($[-0.0077, -0.0020]$). The corresponding yoked contrasts were $+0.0004$ ($[-0.0008, +0.0016]$) and -0.0032 ($[-0.0083, +0.0020]$; approximate 95% CIs). The direct m -only-minus-yoked contrast was $+0.0003$ at A (approximate 95% CI $[-0.0011, +0.0016]$) and -0.0017 at D

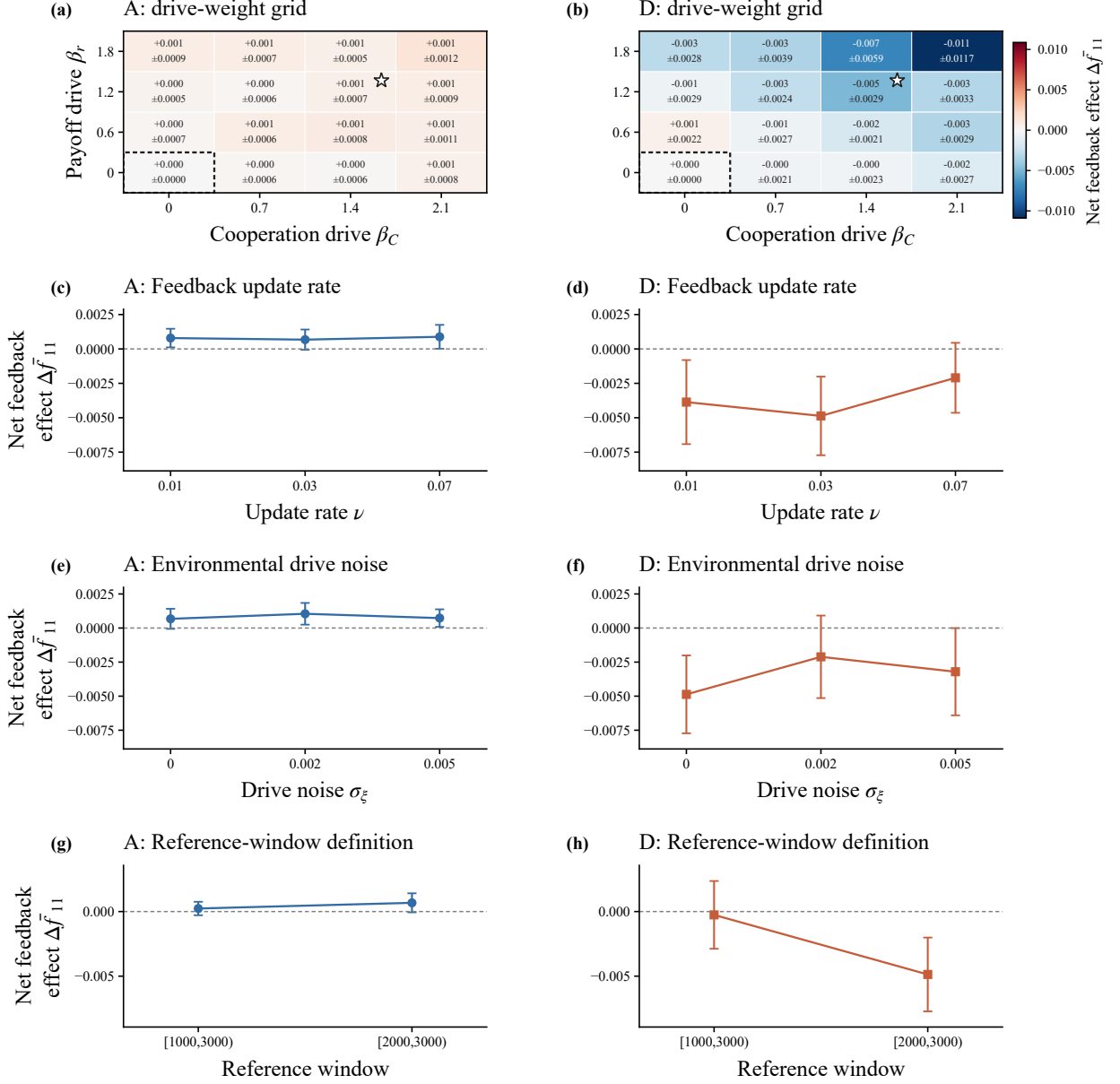


Figure 11: Paired m -only sensitivity at A (left) and D (right). (a,b) Responses over the 4×4 drive-weight grid; dashed outlines mark the zero-drive cells ($\beta_C = \beta_r = 0$), and stars mark $(\beta_C, \beta_r) = (1.4, 1.2)$. (c,d) Environmental update rate. (e,f) Drive noise. (g,h) Reference-window definition. Entries and error bars show means and pointwise 95% Student- t intervals across 20 matched seeds over $[3000, 12000]$.

($[-0.0098, +0.0063]$). The intervals account conservatively for shared donors using alternating groups (Appendix C). Both direct-comparison intervals included zero (Fig. 10a).

Across $N_I = 60, 120, 240, 480$, the fitted crossover width of the static baseline decreased from 0.96 to 0.73 (95% whole-seed bootstrap CIs $[0.84, 1.07]$ and $[0.68, 0.79]$), and the variance peak decreased from 0.0015 to 0.00026 ($[0.00096, 0.0021]$ and $[0.00018, 0.00040]$; Fig. 10d). The shared-seed bootstrap slopes against $\log_2 N_I$ were -0.072 $[-0.11, -0.033]$ for the width and -0.00041 $[-0.00060, -0.00023]$ for the variance peak. Both trends are resolved over the sampled sizes at the fixed 12,000-round horizon: larger populations showed a narrower crossover and lower between-seed variability.

5. Discussion and conclusion

We find an asymmetric trust response to environmental feedback: trust increases near perception-tier boundaries, whereas feedback reinforces learning-driven trust erosion at the high-return operating point D. At $m_0 = 3.3$, changing the perceived environmental tier raised successful trust from 3.2% to 4.3% while both payoff parameters remained fixed. At D, stronger drive and longer horizons deepened the negative response, whereas extending the historical reference window backward attenuated it. The direction of feedback therefore depends on the learners’ states and histories; stronger feedback does not consistently produce higher trust.

We couple a centered environmental update with heterogeneous Q -learning to close the loop between collective behavior, environmental change, and subsequent decisions. Relative to the fixed-payoff trust game, collective outcomes now reshape both material incentives and publicly perceived conditions. Paired branches from a shared learning checkpoint and separate payoff and perception controls identify the trust responses through these two routes. Previous work reported high trust in repeated Q -learning games with alternating roles [29]; our model examines how collective learning changes its own decision environment under fixed roles.

Behavioral heterogeneity links this collective response to individual learning. Risk-seeking investors lead participation, the one-round thresholds explain the RS–RN–RA ordering, and the discount-factor control shows that continuation value advances the RS onset. At the population level, feedback changes successful trust while preserving the pooled two-block NETI sign pattern at B–D. Trust levels can thus change within a retained organization of outcome pairs.

In our tabular learner, changes in the perceived tier activate different Q rows, and the response combines first visits to untrained rows with subsequent learning. This links the perception effect to state representation and learning history. These findings offer an analogy for platforms at different stages of development. In a community’s cold-start phase, perceived public conditions could provide an entry point for building trust. In a mature platform, high returns alone may be insufficient to sustain trust as learning accumulates. This analogy motivates interventions differentiated by trust state and proximity to perception-tier boundaries. For low-trust communities near a boundary, one testable intervention is public feedback that makes improvements in collective cooperation visible. For high-return platforms with declining trust, a complementary test is whether constraining returns to defection or introducing penalties can curb trust erosion. These are hypotheses motivated by the asymmetric model response, to be tested under the information and learning conditions of the platform.

Our results concern two populations with fixed roles, random matching each round, and finite observation horizons. Extending the model to dynamic networks with partner choice and rewiring would establish how evolving relationships alter the payoff and perception responses. Within the present model, trust maintenance depends jointly on material incentives, public state representation, and collective learning history.

CRedit authorship contribution statement

Ruqiang Guo: Conceptualization, Methodology, Writing – original draft.

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Fangfang Wang: Investigation, Formal analysis.

Zhaoyi Hu: Formal analysis, Discussing, & Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Theoretical analysis and derivations

Appendix A.1. NETI properties

Let $X_i(t) \in \mathcal{S}_o := \{00, 01, 10, 11\}$ be investor i 's realized outcome state in round t . For a contiguous analysis window $\mathcal{W}$, we count adjacent pairs

$$N_{ss'} = \sum_{i=1}^{N_I} \sum_{t \in \mathcal{W}} \mathbf{1}[X_i(t-1) = s, X_i(t) = s'], \quad P_o(s, s') = \frac{N_{ss'}}{N_I |\mathcal{W}|}. \quad (\text{A.1})$$

We also compute the pooled current-state frequency $\pi_+(s) = \sum_{s'} P_o(s', s)$, which equals the outcome share used in Eq. (12), and use the symmetric shuffled benchmark $P_e(s, s') = \pi_+(s)\pi_+(s')$ in Eq. (14). The previous-state marginal $\pi_-(s) = \sum_{s'} P_o(s, s')$ can differ from $\pi_+(s)$ because the window replaces its initial boundary state by its final boundary state. For each s this discrepancy is bounded by $1/|\mathcal{W}|$.

For comparison, the empirical conditional transition probability is defined when $\pi_-(s) > 0$ as

$$K_{\mathcal{W}}(s' | s) = \frac{P_o(s, s')}{\pi_-(s)}, \quad (\text{A.2})$$

whereas NETI measures normalized excess joint frequency relative to this shuffled benchmark, retaining the exposure frequency of state s . A positive cell means $P_o > P_e$, a negative cell means $P_o < P_e$, and a zero cell means equality with this finite-window benchmark.

The attainable upper bound depends on the cell marginals. Every joint table obeys

$$P_o(s, s') \leq \min\{\pi_-(s), \pi_+(s')\}.$$

Consequently,

$$\tilde{\kappa}(s, s') \leq \frac{\min\{\pi_-(s), \pi_+(s')\} - \pi_+(s)\pi_+(s')}{1 - \pi_+(s)\pi_+(s')}, \quad (\text{A.3})$$

whenever the denominator is positive. In the limiting degenerate case $P_e = 1$, the formula is undefined and we assign zero by convention. Under equal row and column marginals, Eq. (A.3) is strictly below one for every nondegenerate off-diagonal cell (and equals $\pi(s)/(1 + \pi(s))$ on the diagonal). We therefore compare NETI magnitudes across matched conditions, using seed-level uncertainty and the cell-specific bounds.

We also formed conditional independence benchmarks from the population outcome margins within non-overlapping blocks of L rounds. Let $n_t(s) = \sum_{i=1}^{N_I} \mathbf{1}[X_i(t) = s]$ and let $\mathcal{B}_L$ partition $\mathcal{W} = [3000, 12000)$ into equal blocks. The analytic independent-pairing expectation is

$$P_e^{(L)}(s, s') = \frac{\sum_{B \in \mathcal{B}_L} \left[\sum_{t \in B} n_{t-1}(s) \right] \left[\sum_{t \in B} n_t(s') \right]}{N_I^2 L |\mathcal{W}|}.$$

We used $L = 100$ to preserve each block's lagged and current outcome margins, and $L = 1$ to preserve the exact margins of each adjacent-round pair. For each seed and branch, we substituted $P_e^{(L)}$ for P_e in Eq. (14), retaining the observed full-window P_o .

Appendix A.2. Threshold derivations

For an investor who commits one unit, a trustworthy trustee produces the monetary payoff $r^T = m/2 - 1$ and an untrustworthy one produces $r^U = -1$. For $m \geq 2$, the former

lies on the non-negative branch of Eq. (3). In this myopic benchmark, we denote perceived trustworthiness by $\hat{q}$. The one-round expected subjective utility of investing is

$$U_\theta(1 | \hat{q}) = \hat{q} \left(\frac{m}{2} - 1 \right) - (1 - \hat{q})\lambda_\theta, \quad (\text{A.4})$$

whereas not investing yields zero. Solving $U_\theta(1 | \hat{q}) \geq 0$ gives

$$\hat{q} \geq \tilde{q}_\theta^*(m) = \frac{\lambda_\theta}{m/2 - 1 + \lambda_\theta} = \frac{2\lambda_\theta}{m + 2\lambda_\theta - 2}. \quad (\text{A.5})$$

For $m > 2$, this threshold decreases with m and increases with λ_θ , yielding the ordering in Eq. (15); at $m = 2$ it equals one for every $\lambda_\theta > 0$. The simulated agents choose actions from learned Q -values with constant ε -greedy exploration.

For a trustee with positive inflow $I_j > 0$, let R_j be its reputation before the action. The two possible next reputations are

$$R_j^T(t+1) = (1 - \eta_R)R_j(t) + \eta_R, \quad (\text{A.6})$$

$$R_j^U(t+1) = (1 - \eta_R)R_j(t). \quad (\text{A.7})$$

Although the action-specific changes are $\eta_R(1 - R_j)$ and $-\eta_R R_j$, respectively, their difference is exactly

$$\Delta R_j^T - \Delta R_j^U = \eta_R, \quad (\text{A.8})$$

independent of the current reputation. Meanwhile the immediate monetary advantage of the untrustworthy action is

$$w_j^U - w_j^T = \frac{\rho m I_j}{2}. \quad (\text{A.9})$$

The one-round utility difference between the trustworthy and the untrustworthy action is hence $\lambda_\phi \eta_R - \rho m I_j / 2$, and the trustworthy action is weakly preferred under the myopic comparison when

$$\lambda_\phi \geq \lambda_\phi^*(m, \rho, I_j) = \frac{\rho m I_j}{2\eta_R}. \quad (\text{A.10})$$

When $I_j = 0$, both actions have the same monetary payoff and both set the reputation signal to zero, so they are myopically indifferent; the differential in Eq. (A.8) applies only to positive inflow. Equation (A.10) compares immediate rewards; the learned trustee policy also depends on continuation values, the type-specific discount γ_ϕ and exploration.

Appendix A.3. Linear response of the anchored loop

The drive vanishes at the reference observables, while finite-time baseline drift can supply a nonzero input after the checkpoint. With binary unit investments, the population-mean investor payoff in any round is exactly

$$\bar{r}_I(t) = \left(\frac{m(t)}{2} - 1 \right) f_{11}(t) - f_{10}(t). \quad (\text{A.11})$$

Thus the cooperation-rate term satisfies $\partial E(t+1)/\partial C(t) = \nu\beta_C$ when the payoff input is held fixed. If f_{11} changes at fixed m and f_{10} , the payoff channel contributes

$$\left. \frac{\partial \bar{r}_I}{\partial f_{11}} \right|_{m, f_{10}} = \frac{m}{2} - 1, \quad (\text{A.12})$$

so the combined one-round drive slope along that direction is $\nu[\beta_C + \beta_r(m/2 - 1)]$. Other reallocations of outcome mass give different directional derivatives; for example, converting 10 to 11 at a fixed investment rate changes the payoff by $m/2$ per unit mass. The perturbation direction therefore determines how the two drive terms combine.

For a fixed analysis window, define the ensemble static-baseline responses

$$F_{\mathcal{W}}(m, \rho) = \mathbb{E}_s[\bar{f}_{11,s}], \quad R_{\mathcal{W}}(m, \rho) = \mathbb{E}_s[\bar{r}_{I,s}], \quad (\text{A.13})$$

with the population size, initialization and analysis window held fixed. The local approximation uses smooth proxies for the responses generated by the hard observation bins and ε -greedy action rule. Let $\tilde{F}_{\mathcal{W}}$ and $\tilde{H}_{\mathcal{W}}$ be smooth proxies for the mean f_{11} and f_{10} responses within unchanged payoff and observation tiers. Define the payoff proxy consistently with Eq. (A.11):

$$\tilde{R}_{\mathcal{W}} = \left(\frac{m}{2} - 1\right) \tilde{F}_{\mathcal{W}} - \tilde{H}_{\mathcal{W}}.$$

The formal susceptibilities are $\chi_{C,m}^{\mathcal{W}} = \partial \tilde{F}_{\mathcal{W}} / \partial m$ and $\chi_{r,m}^{\mathcal{W}} = \partial \tilde{R}_{\mathcal{W}} / \partial m$, evaluated at (m_0, ρ_0) within such a region. Equation (8) gives

$$M'(E_0) := \left. \frac{dm}{dE} \right|_{E_0} = k(m_{\max} - m_{\min})z_0(1 - z_0), \quad z_0 = \frac{m_0 - m_{\min}}{m_{\max} - m_{\min}}. \quad (\text{A.14})$$

Within the fixed payoff and observation tiers, the quasi-static scalar linearization reads

$$\delta E(t+1) \simeq [1 - \nu + g_{\mathcal{W}}(m_0, \rho_0)] \delta E(t), \quad (\text{A.15})$$

$$g_{\mathcal{W}} = \nu M'(E_0) (\beta_C \chi_{C,m}^{\mathcal{W}} + \beta_r \chi_{r,m}^{\mathcal{W}}). \quad (\text{A.16})$$

Equation (A.15) governs a small *perturbation* δE about the quasi-static operating point. Writing $G_{\mathcal{W}} = g_{\mathcal{W}}/\nu$, local stability of this scalar map requires $|1 - \nu + \nu G_{\mathcal{W}}| < 1$, or $1 - 2/\nu < G_{\mathcal{W}} < 1$. With a positive multiplier in this range, the perturbation decays monotonically while collective outcomes can drive changes in the operating level $E(t)$.

The payoff susceptibility can be decomposed directly from Eq. (A.11):

$$\chi_{r,m}^{\mathcal{W}} = \frac{1}{2} \tilde{F}_{\mathcal{W}} + \left(\frac{m_0}{2} - 1\right) \chi_{C,m}^{\mathcal{W}} - \frac{\partial \tilde{H}_{\mathcal{W}}}{\partial m}. \quad (\text{A.17})$$

This expression separates the mechanical return effect from behavioral shifts between successful trust (11), untrustworthy outcomes (10), and non-investment.

The simulated $\rho(E)$ map has zero derivative inside each tier and no derivative at a boundary. A crossing can instead be described over a finite interval by $h'_{\rho,\text{eff}} = \Delta \rho / \Delta E$. At fixed m , define response secants over the same endpoints as $\hat{\chi}_{C,\rho}^{\mathcal{W}} = \Delta F_{\mathcal{W}} / \Delta \rho$ and $\hat{\chi}_{r,\rho}^{\mathcal{W}} = \Delta R_{\mathcal{W}} / \Delta \rho$. Their finite-crossing drive coefficient is

$$g_{\rho,\mathcal{W}} = \nu h'_{\rho,\text{eff}} (\beta_C \hat{\chi}_{C,\rho}^{\mathcal{W}} + \beta_r \hat{\chi}_{r,\rho}^{\mathcal{W}}). \quad (\text{A.18})$$

This coefficient describes a finite crossing between tiers; Eqs. (A.15)–(A.16) describe local response within a tier.

Within the smooth approximation, a steeper $\tilde{F}_{\mathcal{W}}$ contributes a larger local gain. The simulated learners also retain the preceding payoff history through their Q -values and visited states. Where a history-dependent linear-response approximation applies, response kernels would therefore replace the static susceptibilities. The scalar gain summarizes a quasi-static response within the fast–slow feedback framework [33, 34].

We construct the finite-difference benchmark for the 0.42 comparison in Section 4.5 from baseline means on the 31-point grid at $\rho_0 = 0.2075$, over $m \in [2.9132, 5.7184]$. Both this grid and the feedback slice use 20 seeds and window $[3000, 12000)$. We define $\hat{F}_W$ by piecewise-linear interpolation of the grid means, with constant extension beyond either endpoint, and compute the slope as

$$\hat{\chi}_{C,m}^W(m_0; h) = \frac{\hat{F}_W(m_0 + h, \rho_0) - \hat{F}_W(m_0 - h, \rho_0)}{2h}, \quad h = 0.25.$$

The comparison retains nominal slice points with $m_0 \geq 3.6$ and distance at least 0.1 from either observation-tier edge. This selection leaves 11 points based on their nominal centers. Intervals centered at $m_0 = 4.5$ and 4.9 cross the upper observation edge, while those at 5.5 , 5.7 and 5.9 use endpoint extension. At each selected point, the benchmark is $\hat{\chi}_{C,m}^W \Delta \bar{m}$. The 11 signed observed-to-benchmark ratios have median 0.42, interquartile range $[0.252, 0.642]$ (linearly interpolated quartiles), and range $[-0.009, 33.125]$. At $m_0 = 5.9$, endpoint extension gives a small benchmark denominator (-0.000117) and a ratio of 33.125; at 5.1 , a near-zero observed response reverses sign relative to the benchmark. We include all 11 points in these ratio summaries.

If an environmental perturbation follows this scalar map with a positive multiplier, we calculate G_{eff} from the time constant τ of an exponential fit:

$$G_{\text{eff}} = \frac{e^{-1/\tau} - 1 + \nu}{\nu}, \quad G_{\text{eff}} \simeq 1 - \frac{1}{\nu\tau}. \quad (\text{A.19})$$

The first expression is the exact discrete-time conversion; the second is its continuous approximation. We infer G_{eff} from the fitted environmental relaxation time, which includes contributions from mechanical payoff feedback. The static-susceptibility expression in Eq. (A.16) defines the separate quasi-static benchmark.

Appendix A.4. Onset estimate

For each parameter setting, q_0 is measured at the specified point $m = 3.0$ by pooling the raw outcome counts over the fixed analysis window,

$$q_0 = \frac{N_{11}}{N_{10} + N_{11}}. \quad (\text{A.20})$$

This point is evaluated independently of the nine-point logistic fit and lies well below its fitted midpoint. Substituting q_0 for $\hat{q}$ in Eq. (A.5) and solving $q_0 = 2\lambda/(m + 2\lambda - 2)$ for m gives Eq. (17). For uncertainty propagation, each bootstrap draw resamples the 20 seed indices as whole units, refits the complete $\bar{f}_{11}(m)$ curve, recomputes q_0 from the corresponding resampled counts, and then evaluates the pooled midpoint's offset from the RS reference. The nine combinations of discount factor and exploration rate in Fig. 8(a) and the five loss-aversion settings in Fig. 8(b) comprise 14 parameter settings, each of which yielded 1,000 valid bootstrap draws. Logistic fits had R^2 values from 0.9366 to 0.9988. The positive offset compares the pooled finite-horizon midpoint with the RS one-round utility zero.

Appendix B. State space and learning details

All continuous state signals are discretized with the same three-level map

$$b_3(x) = \begin{cases} 0, & 0 \leq x < 1/3, \\ 1, & 1/3 \leq x < 2/3, \\ 2, & 2/3 \leq x \leq 1, \end{cases} \quad (\text{B.1})$$

after clipping x to $[0, 1]$. Labels 0, 1, 2 correspond to low, medium, and high. The public environment component bins the branch-specific latent signal $m_{\text{signal}}(E_t)$ through

$$z_{\text{signal}}(t) = \text{clip}\left(\frac{m_{\text{signal}}(E_t) - m_{\min}}{m_{\max} - m_{\min}}, 0, 1\right), \quad b_E(t) = b_3(z_{\text{signal}}(t)). \quad (\text{B.2})$$

All feedback branches use the same signal mapping, $z_{\text{signal}}(t) = \sigma(kE_t)$ by Eq. (8), evaluated along their own environmental trajectories. In the ρ -only branch, the realized payoff multiplication factor remains m_0 , while the public tier is computed from that branch's latent signal. In the static baseline, $m_{\text{signal}} = m_0$ and the corresponding tier is constant.

After investor i is matched to trustee $j(i, t)$, its decision state is

$$s_i(t) = (X_i(t-1), b_3(T_{i,j(i,t)}(t)), b_E(t)). \quad (\text{B.3})$$

Here $X_i(t-1) \in \{00, 01, 10, 11\}$ is the previous public outcome code and $T_{i,j(i,t)}$ is the pair-specific trust weight for the current match. Trustee actions are public in the model even when the matched investor does not invest, so both 00 and 01 are observable outcome codes. There are $4 \times 3 \times 3 = 36$ investor states and two actions, $A_I = \{0, 1\}$, giving a 36×2 table for each investor. Investor type enters through the subjective reward in Eq. (3). Both actions are available in every state for all investor types.

Trustees observe aggregated inflow before acting. With $\mathcal{M}_j(t) = \{i : j(i, t) = j\}$ and $n_j(t) = |\mathcal{M}_j(t)|$, define

$$I_j(t) = \sum_{i \in \mathcal{M}_j(t)} a_i(t), \quad \tilde{I}_j(t) = \begin{cases} I_j(t)/n_j(t), & n_j(t) > 0, \\ 0, & n_j(t) = 0, \end{cases} \quad (\text{B.4})$$

where the stake is one unit, so $\tilde{I}_j$ is the invested share of trustee j 's current matches. Its decision state is

$$s_j(t) = (b_3(R_j(t)), b_3(\tilde{I}_j(t)), a_j(t-1), b_3(\bar{T}_j(t)), b_E(t)), \quad (\text{B.5})$$

$$\bar{T}_j(t) = \frac{1}{N_I} \sum_{i=1}^{N_I} T_{ij}(t).$$

The component cardinalities are $3 \times 3 \times 2 \times 3 \times 3 = 162$. Together with $A_T = \{U, T\}$, each trustee therefore has a 162×2 table.

The continuous memory variables remain in $[0, 1]$ under the updates

$$R_j(t+1) = (1 - \eta_R)R_j(t) + \eta_R \mathbf{1}[I_j(t) > 0] \mathbf{1}[a_j(t) = T], \quad (\text{B.6})$$

$$T_{ij}(t+1) = (1 - \eta_T)T_{ij}(t) + \eta_T \mathbf{1}[a_j(t) = T] \quad \text{if } i \text{ invested with } j, \quad (\text{B.7})$$

and $T_{ij}(t+1) = T_{ij}(t)$ otherwise. Pairwise trust therefore changes through realized investment interactions.

Algorithm 1 gives the round-by-round update order. Both roles use the same environment tier $b_E(t)$ and complete their previous transition once the current decision state is available. The shared round-3000 checkpoint retains both Q -tables, all memories and counters, previous states, actions and rewards, the environmental state, seed-specific references, and random-number streams. Each branch therefore continues from the same learned state, including the transition awaiting its next Q update.

Appendix C. Parameters and run specifications

Table C.3: Simulation settings for Figs. 1–11 and Supplementary Figs. S1–S3. Feedback runs use the centered law, a seed-specific historical reference, and a shared round-3000 checkpoint. Figure intervals use the seed units listed for each run.

Figure	Experiment	(N_I, N_T)	Horizon; window	Seeds	Conditions
Fig. 1	centered- E_0 schematic	—	—	—	—
Fig. 2(a,c)	fixed- ρ_0 feedback comparisons	(240, 120)	12k; [3k, 12k)	0–19	same data as Figs. 5c and S3
Fig. 2(b)	trust trajectories at D	(240, 120)	12k; [3k, 12k)	100–149	same data as Fig. 6l
Fig. 3	baseline grids	(240, 120)	12k; [3k, 12k)	0–19	static baseline
Fig. 4	six- m time series	(240, 120)	12k; [3k, 12k)	0–19	static baseline
Fig. 5	feedback grid	(240, 120)	12k; [3k, 12k)	0–19	baseline, m -only
(a,b,d)					
Fig. 5(c)	fixed- ρ_0 slice	(240, 120)	12k; [3k, 12k)	0–19	baseline, m -only, ρ -only, $m+\rho$
Fig. 6	A–D comparisons	(240, 120)	12k; [3k, 12k)	100–149	baseline, m -only
Fig. 7	mechanism atlas	(240, 120)	12k; [3k, 12k)	0–19	static baseline
Fig. 8	parameter dependence of the fitted onset	(240, 120)	12k; [3k, 12k)	0–19	baseline; bootstrap $B = 1000$
Fig. 9	A–D NETI matrices	(240, 120)	12k; [3k, 12k)	100–149	baseline, m -only; paired
Fig. 10(a)	schedule-yoked	(240, 120)	12k; [3k, 12k)	200–219	baseline, m -only, yoked
Fig. 10(b)	channel comparisons at A and D	(240, 120)	12k; [3k, 12k)	100–149	baseline, m -only, ρ -only, $m+\rho$
Fig. 10	horizon and windows at	(240, 120)	c: 12k, 24k, 36k; [3k, T); e: 12k, refs [1k, 3k) or [2k, 3k); f: 12k, obs. [3k, 12k) or [6k, 12k)	200–219	baseline, m -only
(c,e,f)	A and D				
Fig. 10(d)	baseline finite size	(60, 30) to (480, 240)	12k; [3k, 12k)	0–19	baseline; bootstrap $B = 1000$
Fig. 11	sensitivity grid at A and D	(240, 120)	12k; [3k, 12k)	200–219	baseline, m -only; 4×4 drive, ν , noise, reference window
Fig. S1	baseline horizon diagnostic	(240, 120)	36k; three 3k blocks	0–9	static baseline
Fig. S2	initial-condition control	(240, 120)	12k; [3k, 12k)	0–19	paired low and high initial states
Fig. S3	fixed- ρ_0 channels	(240, 120)	12k; [3k, 12k)	0–19	baseline, m -only, ρ -only, $m+\rho$

We calculated within-seed feedback effects from shared checkpoints and random streams (Table C.3). We also compared the A–D effects between two independent numerical implementations using separate seed ensembles. Pointwise Welch intervals for differences in their m -only-minus-baseline effects all included zero and spanned differences comparable to the feedback effects, especially at C. The source data provide the four estimates and intervals.

For schedule-yoked replay, we index the 20 recipient seeds as $i = 0, \dots, 19$; recipient i receives the schedule from donor $(i + 1) \bmod 20$. A contrast involving replay thus depends on both seeds. We split the cycle into fixed even- and odd-recipient groups, each containing ten disjoint recipient–donor pairs. Assuming independent seed streams, pairs within either group are independent, although the two groups share seeds. For group g , we form a two-sided 97.5% approximate t_9 interval $[L_g, U_g]$ for its mean contrast. We report

$$\hat{\mu} = \frac{\bar{d}_{\text{even}} + \bar{d}_{\text{odd}}}{2}, \quad \left[\frac{L_{\text{even}} + L_{\text{odd}}}{2}, \frac{U_{\text{even}} + U_{\text{odd}}}{2} \right].$$

Bonferroni’s inequality gives at least 95% joint coverage when the marginal intervals attain their nominal coverage; averaging endpoints then covers the overall mean. Coverage remains approximate because each ten-pair interval uses a Student- t approximation. This pointwise procedure uses all 20 contrasts and accounts for shared donors without assuming independence between groups. The m -only-minus-baseline contrast involves no donor and uses the ordinary paired t_{19} interval.

Appendix D. Supplementary numerical experiments

Appendix D.1. Trust dynamics over longer horizons

We extended the baseline runs to 36,000 rounds at three specified points, using 10 seeds. Each 12k, 24k, and 36k checkpoint summarizes the preceding 3,000 rounds within each seed. The convergence gate required an absolute mean 36k-minus-24k change no larger than 0.01 and a paired 95% Student- t interval containing zero.

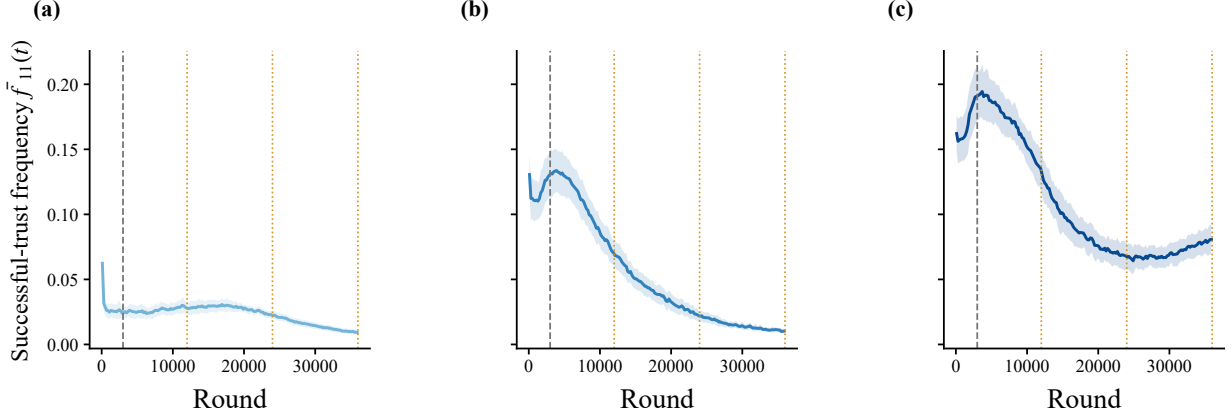


Figure S1: Baseline trust dynamics over 36,000 rounds at $\rho = 0.2075$. Panels (a–c) use $m = 3.2, 4.6$, and 5.6 , respectively. Curves show 200-round block means and two-sided 95% Student- t confidence intervals across 10 static-baseline seeds; the dashed line marks the burn-in cutoff and the dotted lines the checkpoints at 12,000, 24,000, and 36,000 rounds. Successful trust continued to decline at $m = 3.2$ and 4.6 , while $m = 5.6$ showed a late reversal.

Table D.4: Checkpoint means and convergence-gate decisions. The uncertainty on Δ_{36-24} is the paired 95% confidence-interval half-width.

m	12k	24k	36k	Δ_{36-24}	Gate
3.2	0.02802	0.02462	0.01017	-0.01445 ± 0.00280	fail
4.6	0.08233	0.02588	0.01103	-0.01485 ± 0.00380	fail
5.6	0.14789	0.07025	0.07794	$+0.00768 \pm 0.00628$	fail

All three points failed the gate because their paired intervals excluded zero. The first two continued to decline, whereas the high- m point reversed upward late in the run. The 12,000-round estimates therefore describe behavior during ongoing learning.

Appendix D.2. Representative-point and NETI uncertainty

Table D.5 summarizes the paired feedback effects and two NETI-cell contrasts from Figs. 6 and 9. Each NETI cell measures finite-window joint-frequency excess relative to its independent-marginal benchmark.

Appendix D.3. Finite-horizon initial-condition dependence

We paired low- and high-trust initializations across 20 response seeds at $\rho = 0.2075$ and nine specified payoff multiplication factors around the fitted baseline crossover. We set $T_{ij}(0) = R_j(0) = 0.1$ in the low-trust initialization and 0.9 in the high-trust initialization. We set $X_i(-1) = 00$ and $a_j(-1) = U$ in the low-trust initialization, and $X_i(-1) = 11$ and $a_j(-1) = T$ in the high-trust initialization. Both initializations used zero Q -values and the

Table D.5: Feedback effects and paired NETI differences at A–D. Values are means $\pm$ 95% Student- t confidence-interval half-widths over 50 matched seeds and the fixed analysis window.

Point	$\Delta \bar{f}_{11}$	$\Delta \text{NETI } 10 \rightarrow 11$	$\Delta \text{NETI } 11 \rightarrow 11$
A	$+0.000271 \pm 0.000430$	$+0.000016 \pm 0.000033$	$+0.000007 \pm 0.000033$
B	-0.004545 ± 0.002173	-0.000694 ± 0.000462	-0.000553 ± 0.000363
C	-0.003114 ± 0.005335	-0.001583 ± 0.000659	-0.001148 ± 0.000580
D	-0.004022 ± 0.001615	-0.000347 ± 0.000230	-0.000190 ± 0.000230

same seed within each pair. Figure S2 shows the response curves and their paired gap. The high-minus-low gap was positive with a 95% Student- t interval excluding zero at eight of nine sampled points. The interval at $m = 3.9963$ included zero. At the fitted midpoint $m = 4.6168$, the gap was $+0.07000 \pm 0.01569$; it reached $+0.10509 \pm 0.00838$ at the largest sampled m .

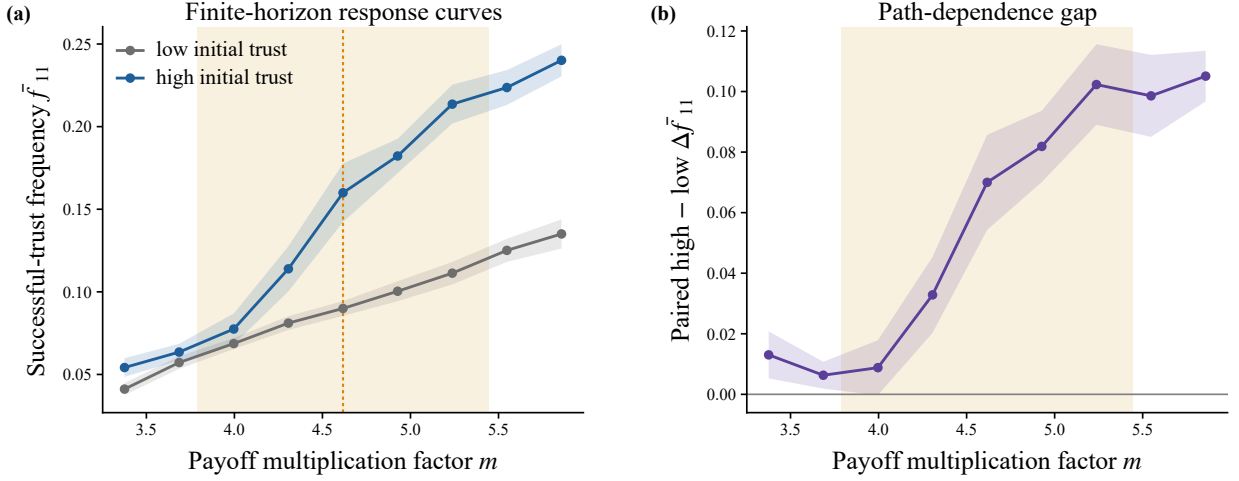


Figure S2: Finite-horizon initial-condition control at $\rho = 0.2075$. (a) Successful-trust frequency for paired low- and high-trust initializations. The shaded region marks the fitted crossover band, and the dotted line marks its midpoint (fitted $m^* = 4.62$, $w = 0.83$ at this ρ). (b) Paired high-minus-low difference. Curves and bands show means and paired 95% Student- t confidence intervals across 20 matched response seeds of the static baseline over the fixed [3000, 12000) analysis window.

These paired trajectories at fixed parameters establish initial-condition dependence over the specified finite window.

Appendix D.4. Centered payoff-channel control

We compared the baseline, m -only, ρ -only, and joint branches across 20 nominal m values at $\rho_0 = 0.2075$, using the same 20 seeds (Fig. S3). Table 2 lists a selection of nine points from this slice. Across the slice, mean m -only changes in f_{11} ranged from -0.0082 to $+0.0103$. The corresponding ranges were -0.00004 to $+0.0108$ for ρ -only and -0.0058 to $+0.0103$ for the joint branch.

Nonzero ρ -only point estimates occurred at sampled $m_0 = 3.3$ – 3.7 and 4.5 – 5.9 , although some intervals included zero. Through the sampled $m_0 = 4.7$ point, this branch retained both nominal payoff parameters and served as a perception control. From $m_0 = 4.9$ upward, realized ρ could also vary, adding payoff changes to the perception response.

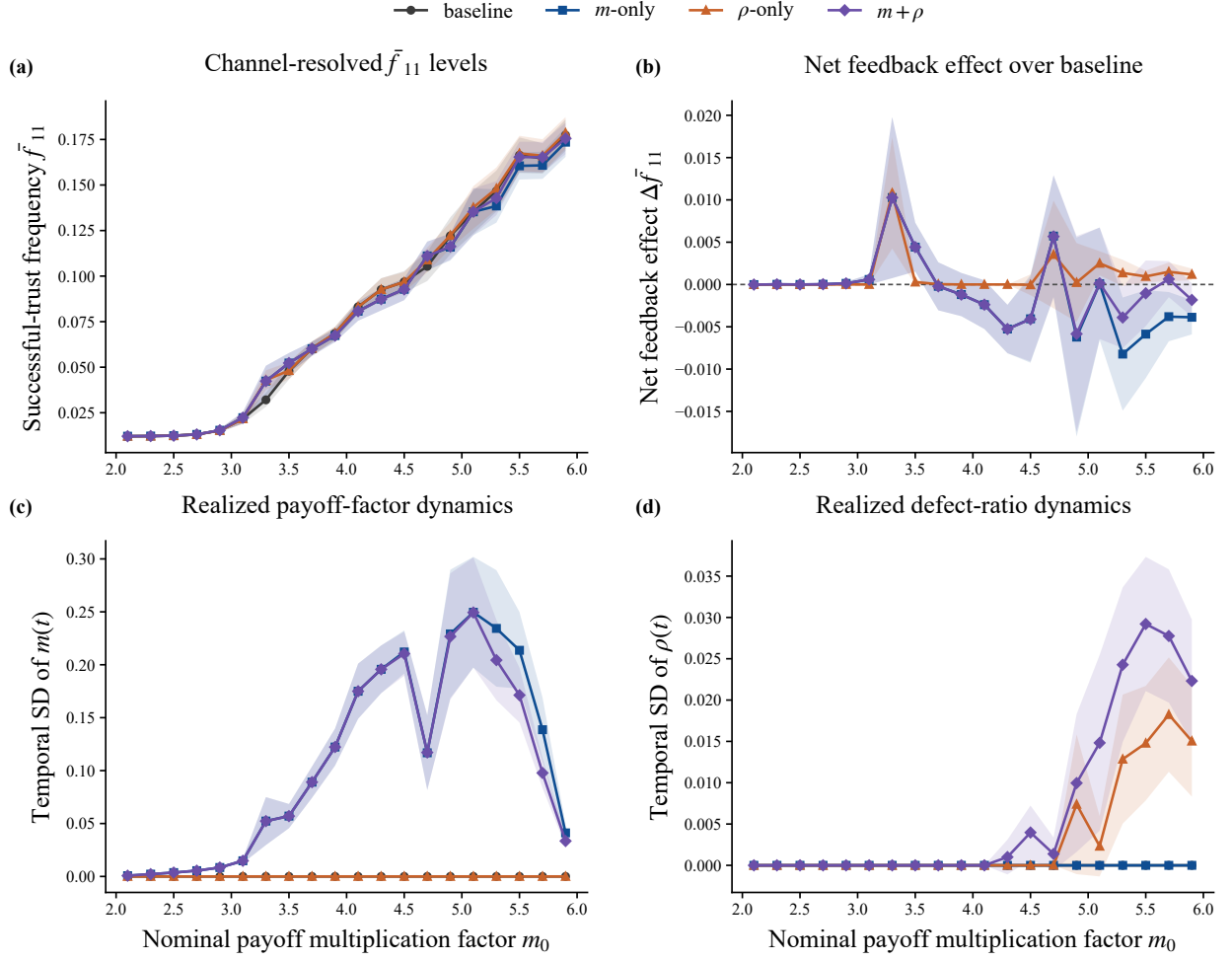


Figure S3: Centered payoff-channel controls at fixed $\rho_0 = 0.2075$. Colors and markers identify the same branches throughout. (a) Successful-trust levels for baseline, m -only, ρ -only, and joint-channel branches. (b) Within-seed branch-minus- baseline changes; the ρ -only branch produces a positive response at $m_0 = 3.3$ while retaining both nominal payoff parameters. The horizontal dashed line in (b) marks zero effect. (c) Temporal standard deviation of realized $m(t)$; it is zero in the baseline and ρ -only branches, where $m(t)$ remains fixed. (d) Temporal standard deviation of realized $\rho(t)$; it is zero in the baseline and m -only branches, where $\rho(t)$ remains fixed. Lines and bands report means and pointwise 95% Student- t confidence intervals across 20 matched seeds.

The payoff factor $m(t)$ remained constant in the baseline and ρ -only branches, while $\rho(t)$ remained constant in the baseline and m -only branches. Each feedback branch followed its own environmental and perceived-state trajectory.

Appendix E. Channel and discretization controls

We tested environmental relaxation, channel effects, continuation value and state discretization. The impulse probe (Appendix E.1) and 2×2 payoff $\times$ perception factorial (Appendix E.2) compare paired groups initialized from the same learned state at round 3000 using common random numbers. The discount-factor (Appendix E.3) and discretization (Appendix E.4) controls change parameters used during learning. For these comparisons, we used matched seeds and separate burn-in periods for each parameter setting, yielding different learned states at round 3000.

Unless stated otherwise, outcome means use $\mathcal{W}_{\text{full}} = [3000, 12000)$. The channel time series also reports $\mathcal{W}_{\text{late}} = [10000, 12000)$, and the impulse fits use the specified post-shock interval. Reused seed identifiers form blocks across operating points and arms. We report each point estimate followed by one standard error. For simple contrasts, this is the sample standard deviation across independent seed blocks divided by $\sqrt{n}$. For fitted or selection-dependent summaries, standard errors come from a delete-one-seed jackknife. We construct pointwise 95% confidence intervals using a Student- t approximation with $n - 1$ degrees of freedom and regard a signed effect as resolved when its interval excludes zero. Contrast definitions and summary results appear below and in Table E.6.

Appendix E.1. Environmental relaxation after a small impulse

Starting from the same learned state and using common random numbers, we imposed positive and negative environmental displacements $\pm\delta E$ ($\delta E = 0.05$) at round $T_0 = 7000$. At the mid-band operating point, no tier edge was crossed, so the dynamic perception bins remained at their nominal value. Both payoff parameters varied with the environment, while the perceived environmental tier remained unchanged. For each of ten seeds (100–109), we formed the symmetrized response $h_s(k) = [m_{+,s}(T_0 + k) - m_{-,s}(T_0 + k)]/2$ and subtracted its mean over $k \in [1800, 2500)$.

We fitted the seed-mean curve to a single exponential using weighted log-linear regression with squared-residual weights $h(k)^2$. The fit used $k \in [5, 200)$ and stopped when the response first fell below $\max(0.03h_{\text{peak}}, 0.0015)$. Here h_{peak} is the mean of the first three response values. The fit required at least 12 points and used $k = 5, \dots, 121$ at the mid-band point. The reported r^2 uses the same squared-residual weights in log space. We obtained the standard error of τ by omitting each seed and repeating the fit.

At the mid-band operating point ($\langle m \rangle \approx 3.86$), the fitted time constant was $\tau = 41.45 \pm 3.51$ rounds (approximate 95% jackknife- t_9 CI [33.51, 49.38]; $r^2 = 0.997$; Fig. S4a). Using the unrounded fitted τ and $\nu = 0.03$, Eq. (A.19) gives $G_{\text{eff}} = 0.205$ in discrete time (approximate 95% CI [0.020, 0.332]) and 0.196 in the continuous approximation ([0.005, 0.325]). We obtain these intervals by monotonically transforming the two endpoints of the τ interval under the scalar model.

The initial response divided by the imposed displacement was 3.92, close to the maximum analytic map slope 4.0. Positive and negative impulse peaks had a magnitude ratio of 1.01. Peak displacement grew approximately linearly with dose at small amplitude (slope 4.21, $r^2 = 0.998$). At $\delta E = 0.45$, it reached 0.81 of the linear extrapolation (Fig. S4c). This saturation is consistent with the bounded map $m(E)$, whose range is (2, 6). As the operating point approached the upper tier edge, the single-exponential fit deteriorated. The fitted τ increased ($41.45 \rightarrow 62.8 \rightarrow 144.6$ rounds as $\langle m \rangle$ moved $3.86 \rightarrow 4.47 \rightarrow 4.59$; Fig. S4b).

This fit characterizes environmental relaxation at the tested operating point, including the mechanical payoff contribution described in Appendix A.3.

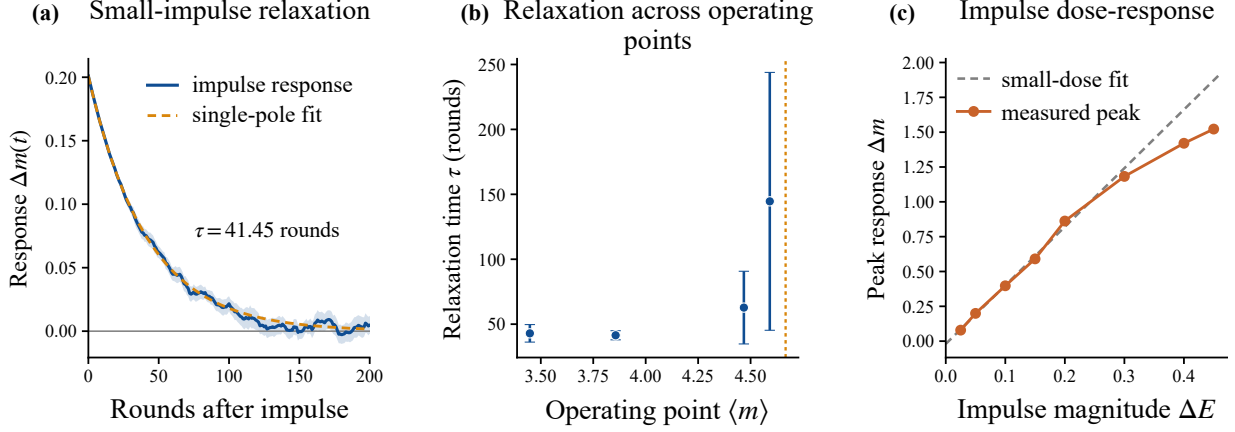


Figure S4: Environmental relaxation and response to different impulse amplitudes. We compared positive and negative perturbations from the same learned state using common random numbers. The environmental tier remained nominal at the mid-band operating point. (a) Symmetrized response of the operating point $m(t)$ to a small $\pm\delta E = 0.05$ impulse at the mid-band operating point $\langle m \rangle \approx 3.86$; the dashed curve is the single-pole fit $Ae^{-t/\tau}$ ($\tau = 41.45$ rounds, $r^2 = 0.997$). The line is the offset-corrected ten-seed mean and the band its ± 1 standard error. (b) Fitted relaxation time τ versus operating point (markers with ± 1 delete-one-seed jackknife standard error); the orange dotted line marks the boundary between the middle and high perception tiers at $m = 14/3 \simeq 4.667$, near which the single-exponential fit deteriorated. (c) Peak displacement versus impulse magnitude with a small-signal linear fit (slope 4.21, $r^2 = 0.998$); the response saturates gently at larger dose. All contrasts are within-seed paired.

Appendix E.2. Factorial payoff and perception controls

We compared four combinations of dynamic or fixed payoff parameters and perceived environmental tiers in a 2×2 factorial design. Each comparison started from the same learned state and used common random numbers, with payoff factor P and perception factor S . The payoff-on setting varied both m and ρ ; the main-text m -only branch varied only the payoff multiplication factor. The four combinations c were defined as follows:

Cell	Realized payoff parameters	Perceived environmental tier
P_+S_+	$m_{\text{signal}}(E_t^c), \rho_{\text{dyn}}(E_t^c)$	dynamic, $b_E(E_t^c)$
P_+S_-	$m_{\text{signal}}(E_t^c), \rho_{\text{dyn}}(E_t^c)$	fixed at its nominal value
P_-S_+	m_0, ρ_0	dynamic, $b_E(E_t^c)$
P_-S_-	m_0, ρ_0	fixed at its nominal value

Each cell updates its own E_t^c from C_t^c and $\bar{r}_{I,t}^c$ using Eq. (11), with shared initial state and reference values. In P_-S_- , the latent environment continues to update while both routes to the agents remain closed. Common random numbers preserve the pairing but allow realized environment paths to diverge. We compared successful trust across the four combinations to estimate payoff and perception effects and their interaction, including the subsequent environmental changes within each combination.

The operating points are $m_0 = 3.3, 4.0$ and 4.7 , each with seeds 100–109. Let $Y_{ab,s,j}^W$ be the successful trust frequency in cell (a,b) , seed s and operating point j , averaged over window W ; in the following expressions the seed and point indices are suppressed:

$$\begin{aligned}
d_P^W &= \frac{1}{2}[(Y_{++}^W - Y_{-+}^W) + (Y_{+-}^W - Y_{--}^W)], \\
d_S^W &= \frac{1}{2}[(Y_{++}^W - Y_{+-}^W) + (Y_{-+}^W - Y_{--}^W)], \\
i^W &= Y_{++}^W - Y_{+-}^W - Y_{-+}^W + Y_{--}^W.
\end{aligned}$$

For the two boundary points $j \in \{L, U\}$, we distinguish the signed average and the mean absolute response:

$$u_s^W = \frac{1}{2}(d_{S,s,L}^W + d_{S,s,U}^W), \quad a_s^W = \frac{1}{2}(|d_{S,s,L}^W| + |d_{S,s,U}^W|),$$

$$\hat{\mu}_W = \frac{1}{10} \sum_s u_s^W, \quad \hat{A}_W = \frac{1}{10} \sum_s a_s^W.$$

For the interaction averaged over the three fixed operating points, we first form $\bar{i}_s^W = \frac{1}{3} \sum_j i_{s,j}^W$ and then average over seeds. Standard errors and t_9 intervals use these ten seed-block values, preserving dependence among the 20 or 30 seed-point cells.

At the mid-band point, the payoff main effect was small but resolved, $\Delta \bar{f}_{11} = -0.0030 \pm 0.0011$. Dynamic and frozen cells had identical successful-trust trajectories at both payoff settings, giving zero perception and interaction contrasts (Fig. S5c).

At the two boundary points, the full-window mean absolute perception effect was $\hat{A}_{\text{full}} = 0.0098 \pm 0.0016$, whereas the signed mean was $\hat{\mu}_{\text{full}} = +0.0021 \pm 0.0034$. The per-point signed effects were $+0.0007 \pm 0.0018$ at the lower edge and $+0.0036 \pm 0.0062$ at the upper, with both 95% intervals including zero. The time-resolved contrasts changed sign. At the upper boundary, the initially negative contrast became positive in $\mathcal{W}_{\text{late}}$, $+0.0310 \pm 0.0063$ (Fig. S5b). For this late window, the corresponding two-boundary absolute mean was $\hat{A}_{\text{late}} = 0.0191 \pm 0.0025$.

The full-window payoff main effects were $+0.0030 \pm 0.0017$ at the lower boundary, -0.0030 ± 0.0011 mid-band and -0.0050 ± 0.0019 at the upper boundary. The lower-boundary 95% interval $[-0.0008, +0.0069]$ included zero; the other two excluded zero. The interaction averaged across all three points was $+0.0021 \pm 0.0021$ (95% CI $[-0.0026, +0.0067]$, ten seed blocks). Its interval spanned zero and included magnitudes comparable to the main effects.

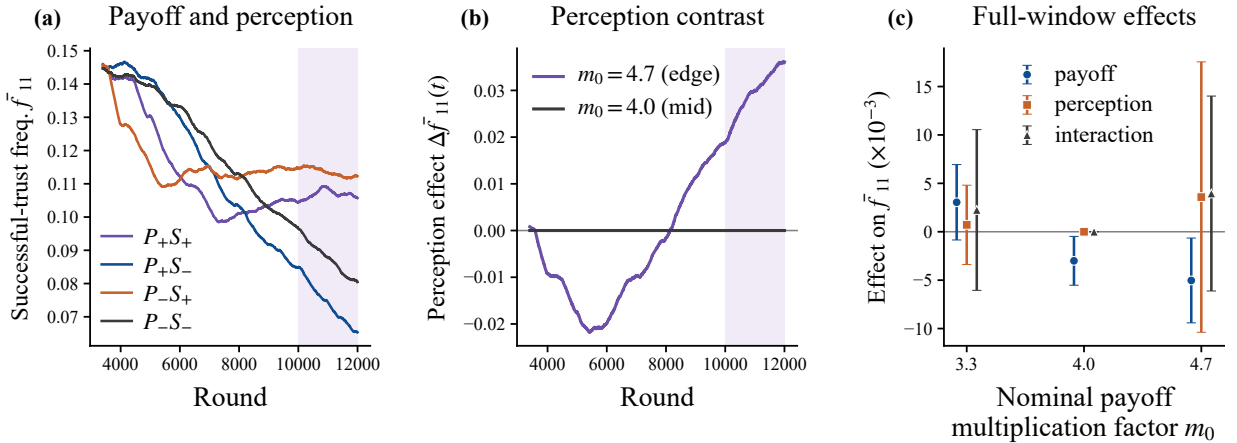


Figure S5: Payoff and perception effects in a 2×2 factorial comparison initialized from the same learned state using common random numbers and ten matched seeds per point. P_+ and P_- denote conditions with environmental payoff feedback enabled and disabled, respectively; S_+ and S_- denote dynamic and fixed perceived environmental tiers, respectively. The P_+ cells vary both m and ρ , and each cell evolves its own environment. (a) Four-cell successful trust frequency $\bar{f}_{11}$ (400-round rolling mean of the seed average) at the upper edge $m_0 = 4.7$; shading in (a,b) marks $\mathcal{W}_{\text{late}} = [10000, 12000]$. (b) Perception contrast $\Delta \bar{f}_{11}(t) = \frac{1}{2}[(P_+S_+ - P_+S_-) + (P_-S_+ - P_-S_-)]$: a build-up near the upper boundary ($m_0 = 4.7$) versus an exactly zero line in the mid-band ($m_0 = 4.0$). (c) Payoff and perception main effects and their interaction over $\mathcal{W}_{\text{full}} = [3000, 12000]$ versus m_0 (markers with pointwise 95% t_9 intervals over ten seeds). At $m_0 = 4.0$, the observed perception main effect and interaction are both exactly zero.

Appendix E.3. Continuation value and the RS fitted midpoint

Figure 8 compares the pooled three-type midpoint with the RS one-round reference in Eq. (17). We tested the role of continuation value by comparing the learner with $\gamma_{\text{inv}} = 0.9$ against a myopic control ($\gamma_{\text{inv}} = 0$), holding the other parameters fixed. We fitted the risk-seeking type's $\bar{f}_{11}(m)$ response curves, which yielded $r^2 \approx 0.98\text{--}0.99$.

For each discount factor, we fitted the risk-seeking response across the same m grid. We estimated uncertainty by deleting each of the six matched seeds from both discount-factor settings and all grid points. Each deletion repeated the logistic fits and the q_0 estimate at $\gamma_{\text{inv}} = 0.9$ at $m = 3.0$. This jackknife propagated covariance between the fitted midpoints and the common RS one-round reference, $m_1^* = 3.035 \pm 0.031$.

With the investor continuation value removed, the fitted RS midpoint was 4.24 ± 0.08 , a gap of $+1.21 \pm 0.08$ above the one-round reference estimated at $\gamma_{\text{inv}} = 0.9$. Both groups continued to update their Q -tables throughout the run. Restoring continuation value lowered the RS midpoint to 3.61 ± 0.04 , a paired shift of -0.63 ± 0.08 (95% approximate t_5 CI $[-0.85, -0.41]$). This paired discount-factor contrast shows that continuation value advances the RS crossover.

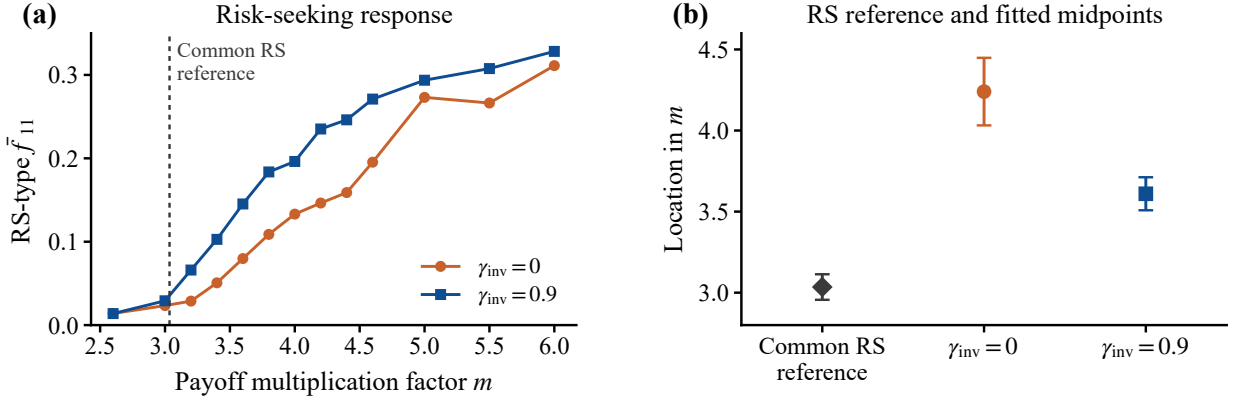


Figure S6: Dependence of the RS fitted midpoint on the discount factor. We varied only the investor discount factor γ_{inv} and used six matched seeds, with separate burn-in periods for each setting. (a) RS-type $\bar{f}_{11}(m)$ for the myopic ($\gamma_{\text{inv}} = 0$) and $\gamma_{\text{inv}} = 0.9$ learners; the dashed line marks the common RS one-round reference ($m_1^* = 3.03$). (b) The common reference, estimated at $\gamma_{\text{inv}} = 0.9$, and the two fitted midpoints. Error bars are approximate pointwise 95% t_5 intervals from a matched-seed jackknife.

Appendix E.4. Sensitivity to the common discretization thresholds

We compared perception-switch effects under two sets of common discretization thresholds. The environmental edges satisfy $m_{\text{edge}} = 2 + 4\kappa$. The schemes used the default thresholds $\kappa = (\frac{1}{3}, \frac{2}{3})$ (edges at 3.333 and 4.667) and alternative thresholds $\kappa = (0.25, 0.75)$ (edges at 3.000 and 5.000). For each threshold scheme, m_0 , and seed, we started from the same learned state and compared dynamic and fixed perceived environmental tiers. Both m and ρ varied with the environment in each comparison. The common κ also defines the $\rho(E)$ tiers. Changing κ moves their switching boundaries from $E - E_0 = \pm \ln 2/k$ to $\pm \ln 3/k$, while retaining the three ρ values. For each matched pair, we measured the successful-trust contrast $dPE = \bar{f}_{11}(\text{dynamic}) - \bar{f}_{11}(\text{frozen})$.

The paired cells shared κ , the payoff update rule, and the starting values of all other κ -binned features (reputation, inflow, and trust). The perception switch was the only difference imposed within each pair. While the dynamic tier remains nominal, both cells present the

same states to the learners. A tier difference can change actions, subsequent payoffs and environmental paths. Thus dPE measures the total closed-loop perception-switch effect within a scheme. Across schemes, changing κ jointly alters state discretization, the ρ -switching rule and the independently generated learning histories.

At $m_0 = 3.333$, near its lower boundary, the $\kappa = (\frac{1}{3}, \frac{2}{3})$ scheme gave $dPE = +0.0254 \pm 0.0091$ (95% t_5 CI $[+0.0021, +0.0488]$). The changed scheme produced identical stored successful-trust trajectories in all six seed pairs. At $m_0 = 3.000$, the changed scheme yielded a nonzero contrast and the $\kappa = (\frac{1}{3}, \frac{2}{3})$ scheme an observed zero. The upper-boundary comparison was less distinct: at $m_0 = 5.000$, the changed scheme had a smaller absolute point estimate and a wide interval. The $\kappa = (\frac{1}{3}, \frac{2}{3})$ scheme gave a larger absolute estimate (Fig. S7).

For the proximity summary, each run's operating point was the frozen-perception branch's full-window mean m_{signal} . Absolute seed-level contrasts pooled across schemes and nominal points averaged 0.0168 ± 0.0032 at distance at most 0.10 from the nearest edge (17 seed-point contrasts). They averaged 0.0056 ± 0.0029 at distance at least 0.30 (56 contrasts). Standard errors deleted each of the six seeds across both schemes and all points, preserving dependence from seed reuse.

The correlation of absolute contrasts with proximity to the scheme's own edges was $+0.24$, compared with $+0.01$ for the other scheme's edges. The change between nonzero and zero perception contrasts near the lower edges demonstrates sensitivity to the common discretization thresholds.

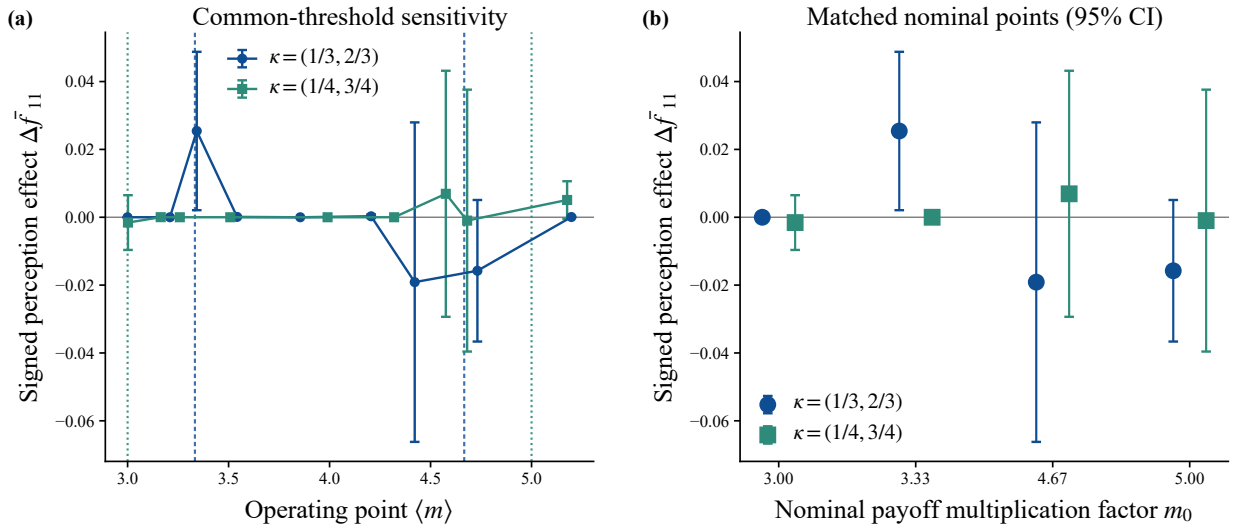


Figure S7: Perception responses under two sets of common discretization thresholds. We used six matched seeds per point and separate burn-in periods for each threshold scheme κ . (a) Signed mean perception contrast dPE versus operating point for the $\kappa = (\frac{1}{3}, \frac{2}{3})$ scheme (edges at 3.33 and 4.67, dashed vertical lines) and the changed scheme (edges at 3.00 and 5.00, dotted vertical lines). (b) Comparisons at matched nominal m_0 , including the nonzero-versus-zero contrast at $m_0 = 3.333$ and the less distinct upper-boundary results at $m_0 = 5.000$. Error bars in both panels are pointwise 95% t_5 intervals. Both m - and ρ -feedback are active. The common κ changes environmental and other state bins as well as the ρ -switching boundaries; the cross-scheme comparison measures their joint sensitivity.

Table E.6: Summary of the channel and discretization controls. Uncertainties are one standard error across seed blocks or from a delete-one-seed jackknife, as specified in the text. A dagger (†) marks signed effects whose pointwise 95% Student- t interval excludes zero. “Exact 0” denotes identical successful-trust trajectories. Outcome windows are $\mathcal{W}_{\text{full}} = [3000, 12000)$ and $\mathcal{W}_{\text{late}} = [10000, 12000)$.

Control	Quantity	Value
Environmental impulse (Appendix E.1)	mid-band relaxation time τ	41.45 ± 3.51 rounds
	scalar G_{eff} in discrete time	0.205
	scalar G_{eff} in the continuous approximation	0.196
	initial response normalized by δE (maximum map slope 4.0)	3.92
	dose-response linear slope	4.21 ($r^2 = 0.998$)
Channel 2×2 (Appendix E.2)	full-window payoff effect, $m_0 = 4.0$	$-0.0030 \pm 0.0011^\dagger$
	perception main + interaction, mid-band	exact 0
	full-window boundary magnitude $\hat{A}_{\text{full}}$	0.0098 ± 0.0016
	full-window boundary signed mean $\hat{\mu}_{\text{full}}$	$+0.0021 \pm 0.0034$
	late-window boundary magnitude $\hat{A}_{\text{late}}$	0.0191 ± 0.0025
	late-window signed perception effect, $m_0 = 4.7$	$+0.0310 \pm 0.0063^\dagger$
	full-window interaction, three-point mean	$+0.0021 \pm 0.0021$
RS midpoint (Appendix E.3)	common one-round reference m_1^*	3.035 ± 0.031
	myopic ($\gamma_{\text{inv}}=0$) midpoint	4.24 ± 0.08
	myopic midpoint minus common reference	$+1.21 \pm 0.08^\dagger$
	$\gamma_{\text{inv}}=0.9$ midpoint	3.61 ± 0.04
	continuation shift, $\gamma_{\text{inv}}:0 \rightarrow 0.9$	$-0.63 \pm 0.08^\dagger$
Discretization (Appendix E.4)	dPE at $m_0=3.333$, $\kappa=(\frac{1}{3}, \frac{2}{3})$	$+0.0254 \pm 0.0091^\dagger$
	same nominal point, changed scheme	exact 0
	mean $ dPE $, distance ≤ 0.10	0.0168 ± 0.0032
	mean $ dPE $, distance ≥ 0.30	0.0056 ± 0.0029
	corr($ dPE $, proximity): own vs other	$+0.24$ vs $+0.01$

References

- [1] Berg J, Dickhaut J, McCabe K. Trust, reciprocity, and social history. *Games Econom Behav* 1995;10(1):122–42.
- [2] Johnson ND, Mislin AA. Trust games: a meta-analysis. *J Econ Psychol* 2011;32(5):865–89.
- [3] Ashraf N, Bohnet I, Piankov N. Decomposing trust and trustworthiness. *Exp Econ* 2006;9(3):193–208.
- [4] Fehr E. On the economics and biology of trust. *J Eur Econ Assoc* 2009;7(2–3):235–66.
- [5] Perc M, Jordan JJ, Rand DG, Wang Z, Boccaletti S, Szolnoki A. Statistical physics of human cooperation. *Phys Rep* 2017;687:1–51.
- [6] Szolnoki A, Chen X. Gradual learning supports cooperation in spatial prisoner’s dilemma game. *Chaos Solitons Fractals* 2020;130:109447.
- [7] Szolnoki A, Perc M. The self-organizing impact of averaged payoffs on the evolution of cooperation. *New J Phys* 2021;23(6):063068.
- [8] Szolnoki A, Chen X. Tactical cooperation of defectors in a multi-stage public goods game. *Chaos Solitons Fractals* 2022;155:111696.

- [9] Capraro V, Perc M. Mathematical foundations of moral preferences. *J R Soc Interface* 2021;18(175):20200880.
- [10] Abbass H, Greenwood G, Petraki E. The N -player trust game and its replicator dynamics. *IEEE Trans Evol Comput* 2016;20(3):470–4.
- [11] Liu Y, Wang L, Guo R, Hua S, Liu L, Zhang L, Han TA. Evolution of trust in the N -player trust game with transformation incentive mechanism. *J R Soc Interface* 2025;22(224):20240726.
- [12] Xia C, Hu Z, Zhao D. Costly reputation building still promotes the collective trust within the networked population. *New J Phys* 2022;24(8):083041.
- [13] Zhou C, Zhu Y, Xia C, Chica M. Evolutionary dynamics of trust in hierarchical populations with varying investment strategies. *J R Soc Interface* 2025;22(225):20240734.
- [14] Kumar A, Capraro V, Perc M. The evolution of trust and trustworthiness. *J R Soc Interface* 2020;17(169):20200491.
- [15] Guo R, Liu L, Liu Y, Zhang L. Evolution of trust in a hierarchical population with different investors based on investment behavioral theory. *Chaos Solitons Fractals* 2023;176:114078.
- [16] Liu Y, Liu L, Guo R, Zhang L. N -player repeated evolutionary trust game under government management. *Chaos Solitons Fractals* 2023;175:114031.
- [17] Sun K, Liu Y, Chen X, Szolnoki A. Evolution of trust in a hierarchical population with punishing investors. *Chaos Solitons Fractals* 2022;162:112413.
- [18] Shang L, Zhang T, Chang M. Evolutionary dynamics in networked trust games with diverse investment patterns. *Europhys Lett* 2023;141(2):22002.
- [19] Liu L, Chen X. Conditional investment strategy in evolutionary trust games with repeated group interactions. *Inform Sci* 2022;609:1694–705.
- [20] Chiong R, Dhakal S, Chaston T, Chica M. Evolution of trust in the sharing economy with fixed provider and consumer roles under different host network structures. *Knowl-Based Syst* 2022;236:107496.
- [21] Watkins CJCH, Dayan P. Q-learning. *Mach Learn* 1992;8(3):279–92.
- [22] Tuyls K, 't Hoen PJ, Vanschoenwinkel B. An evolutionary dynamical analysis of multi-agent learning in iterated games. *Auton Agents Multi-Agent Syst* 2006;12(1):115–53.
- [23] Zheng G, Zhang J, Deng S, Cai W, Chen L. Evolution of cooperation in the public goods game with Q-learning. *Chaos Solitons Fractals* 2024;188:115568.
- [24] Li BY, Zhang ZN, Zheng GZ, Cai CR, Zhang JQ, Chen L. Cooperation in public goods games: leveraging other-regarding reinforcement learning on hypergraphs. *Phys Rev E* 2025;111(1):014304.
- [25] Xie K, Szolnoki A. Reputation in public goods cooperation under double Q-learning protocol. *Chaos Solitons Fractals* 2025;196:116398.

- [26] Lin J, Long P, Liang J, Dai Q, Li H, Yang J. The coevolution of cooperation: integrating Q-learning and occasional social interactions in evolutionary games. *Chaos Solitons Fractals* 2025;194:116165.
- [27] Lv S, Li J, Zhao C. Reinforcement learning in spatial public goods games with environmental feedbacks. *Chaos Solitons Fractals* 2025;195:116296.
- [28] Yang Y, Zhao D, Wang J. Evolution of cooperation in spatial public goods games driven by reinforcement learning and environmental feedback. *Chaos Solitons Fractals* 2025;199:116592.
- [29] Zheng G, Zhang J, Zhang J, Cai W, Chen L. Decoding trust: a reinforcement learning perspective. *New J Phys* 2024;26(5):053041.
- [30] Zhu Y, Xing B, Xia C. Q-learning update with second-order reputation promotes the evolution of trust within structured populations. *Chaos Solitons Fractals* 2025;199:116653.
- [31] Hu Z, Zhu Y, Zhao D, Xia C. The higher-order networked N -player trust game driven by reputation and reinforcement learning. *Chaos Solitons Fractals* 2026;202:117623.
- [32] Perc M, Szolnoki A. Coevolutionary games—a mini review. *Biosystems* 2010;99(2):109–25.
- [33] Weitz JS, Eksin C, Paarporn K, Brown SP, Ratcliff WC. An oscillating tragedy of the commons in replicator dynamics with game-environment feedback. *Proc Natl Acad Sci USA* 2016;113(47):E7518–25.
- [34] Tilman AR, Plotkin JB, Akçay E. Evolutionary games with environmental feedbacks. *Nat Commun* 2020;11(1):915.
- [35] Zhang Z, Yang C, Han W, Zhao D, Xia C. Co-evolutionary dynamics of asymmetric N -player trust game with margin-driven environmental feedback. *Chaos* 2025;35(12):123121.
- [36] Liu Y, Wang L, Guo R, Hua S, Liu L, Zhang L. Evolution of trust in N -player trust games with loss assessment. *Chaos* 2024;34(9):093101.
- [37] Jiang X, Jia W. Reputation-based nonlinear investment promotes cooperation in networked N -player trust games. *Chaos* 2026;36(6):063130.
- [38] Kahneman D, Tversky A. Prospect theory: an analysis of decision under risk. *Econometrica* 1979;47(2):263–91.
- [39] Tversky A, Kahneman D. Advances in prospect theory: cumulative representation of uncertainty. *J Risk Uncertain* 1992;5(4):297–323.
- [40] Chica M, Chiong R, Kirley M, Ishibuchi H. A networked N -player trust game and its evolutionary dynamics. *IEEE Trans Evol Comput* 2018;22(6):866–78.
- [41] Cohen J. A coefficient of agreement for nominal scales. *Educ Psychol Meas* 1960;20(1):37–46.